Computational Sociology

Introduction to Large Language Models

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October 30, 2026

Plan

1. Course updates
2. Language models and transformers
3. Large language models
4. Post-training, reasoning, and agents
5. Text classification with LLMs
6. Language understanding, meaning, and interpretability
7. Challenges

Course updates

Project progress report

- Progress report due next Friday, November 6
  - 2–3-page double-spaced update on your final project (see last week's slides)
- Submit PDF via email

Course updates

Homework 3

- Homework 3 released after class today
  - Supervised machine learning
  - Supervised text classification
  - Large language models
- Due Friday, November 13

Part I: Language models and transformers

Language models

Review

- Language models predict the likelihood of a sequence of words
  - Applications include auto-completion, speech recognition, and more
- In the bigram model, the probability of w_k only depends on the previous word, w_{k-1}

$$P(w_{1:n}) \approx \prod_{k=1}^{n} P(w_k|w_{k-1})$$

- n-gram language models generalize this to longer sequences of words

Language models

Limitations of n-gram language models

1. Insufficient data to sufficiently model language generation
2. Complex models become intractable to compute
3. Limited information on word order

Language models

Advances in neural language models

- Over the past decade language models have developed due to three factors:
  1. Availability of large text corpora
  2. Advances in computer processing, notably Graphics Processing Units (GPUs)
  3. Innovations in neural network architecture

Language models

Word2vec

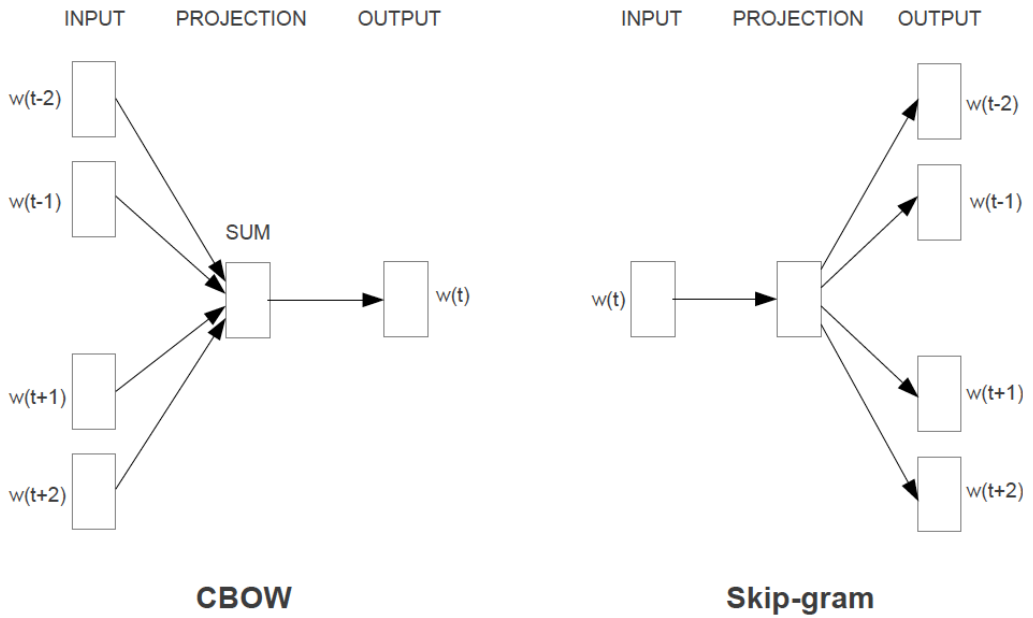


Figure 1: New model architectures. The CBOW architecture predicts the current word based on the context, and the Skip-gram predicts surrounding words given the current word.

Mikolov et al. 2013

Language models

From static to contextualized embeddings

- Word2vec learns a single vector for each word, averaging across its different uses
  - e.g. “crane” as a bird, construction equipment, or movement
- Contextualized embeddings allow the vector for a word to vary according to its context
- The *attention* mechanism makes this possible

Attention and the transformer architecture

Attention Is All You Need

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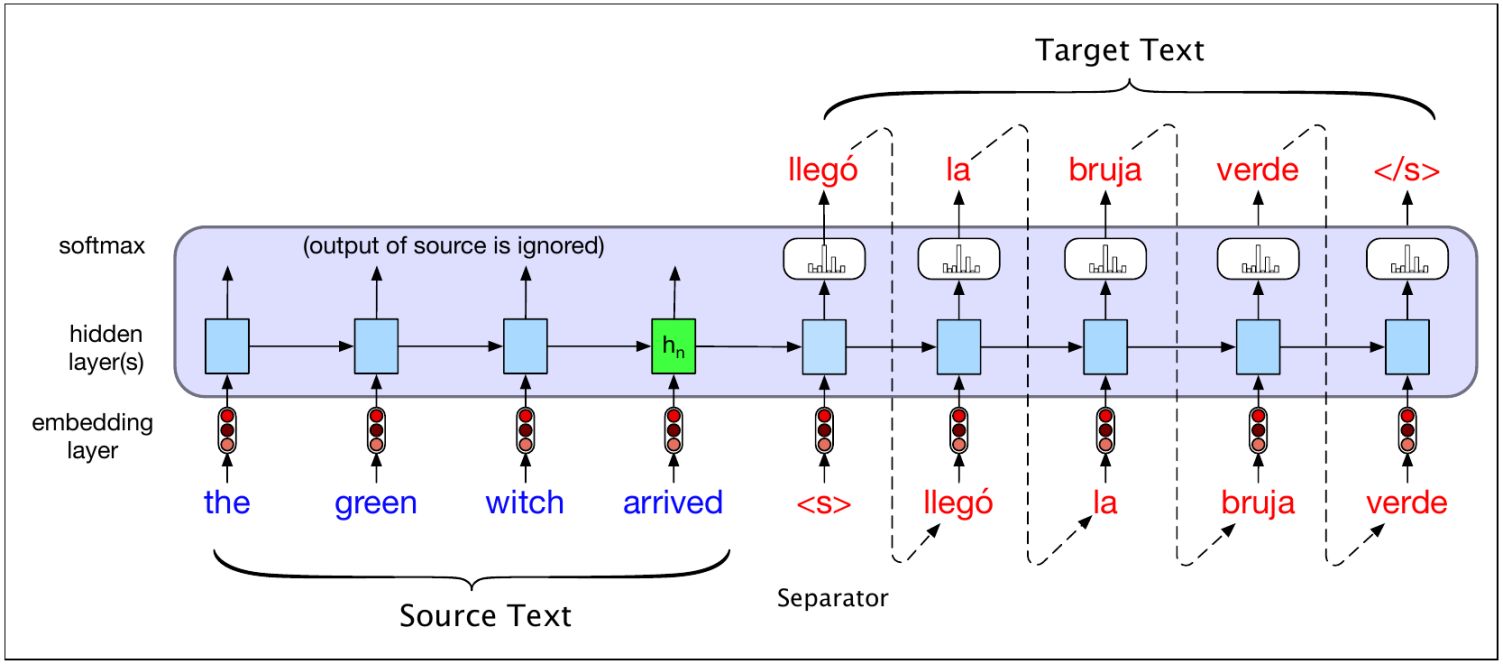
Illia Polosukhin[*] [‡]
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Abstract

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-to-German translation task, improving over the existing best results, including ensembles, by over 2 BLEU. On the WMT 2014 English-to-French translation task, our model establishes a new single-model state-of-the-art BLEU score of 41.0 after training for 3.5 days on eight GPUs, a small fraction of the training costs of the best models from the literature.

Attention and the transformer architecture

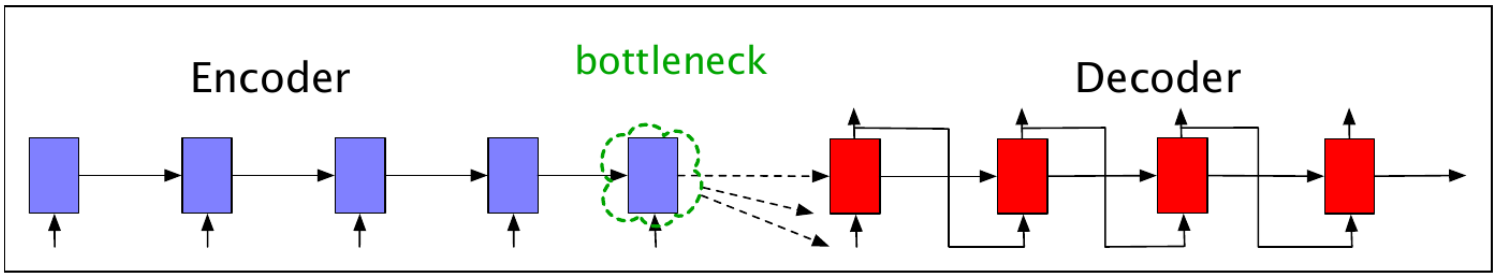
Encoders and decoders



The decoder generates the target text using the encoder's final hidden state, here producing a translation. Jurafsky and Martin 2026

Attention and the transformer architecture

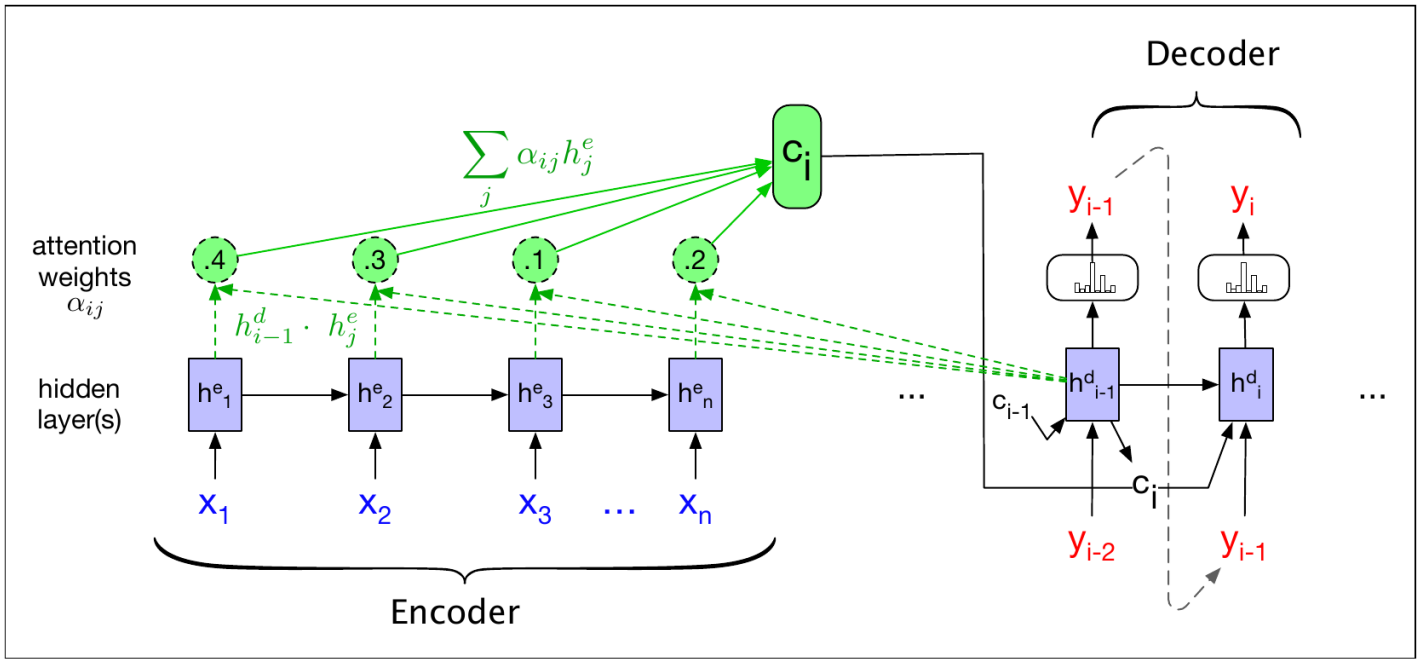
The bottleneck



All information in a standard encoder-decoder passes through the final state. Jurafsky and Martin 2026

Attention and the transformer architecture

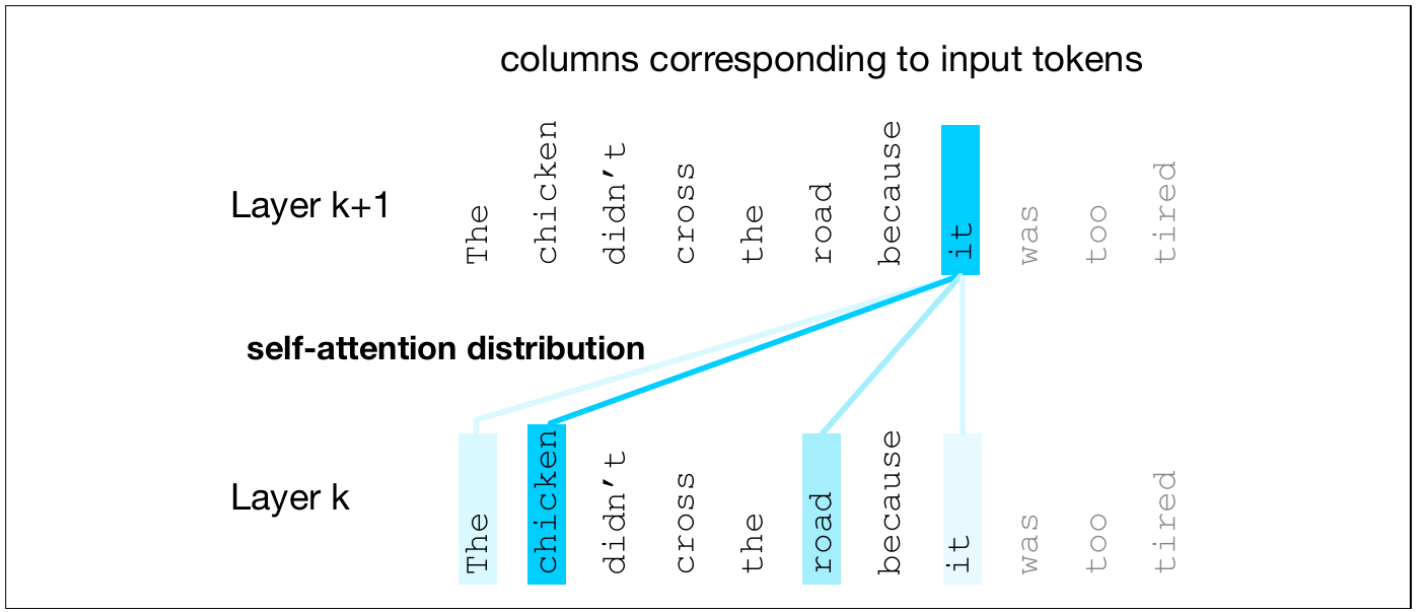
Adding attention



Attention allows the context for each step of the decoder to be represented as a weighted sum of all the encoder's hidden states. Jurafsky and Martin 2026

Attention and the transformer architecture

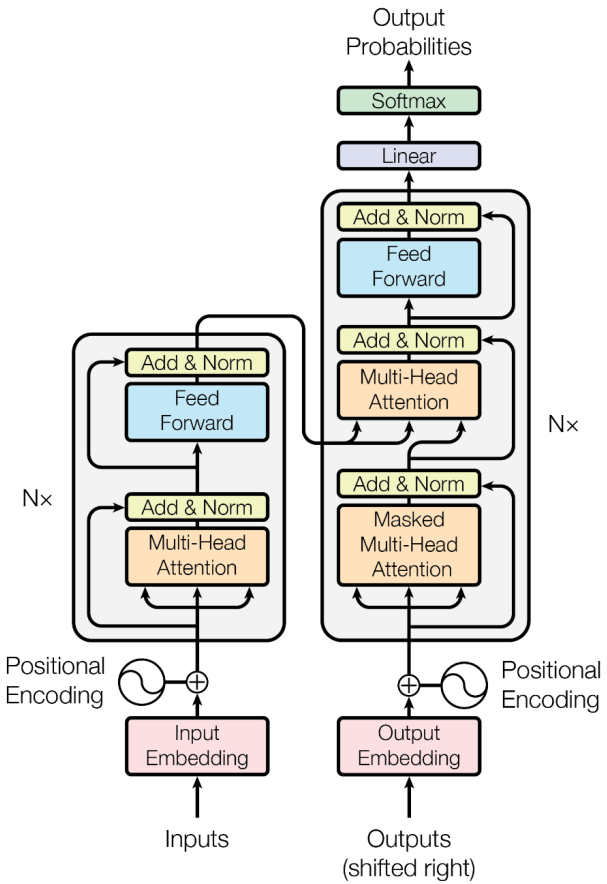
Attending to a token



When computing the representation for "it," the model attends most to "chicken" and "road." Jurafsky and Martin 2026

Attention and the transformer architecture

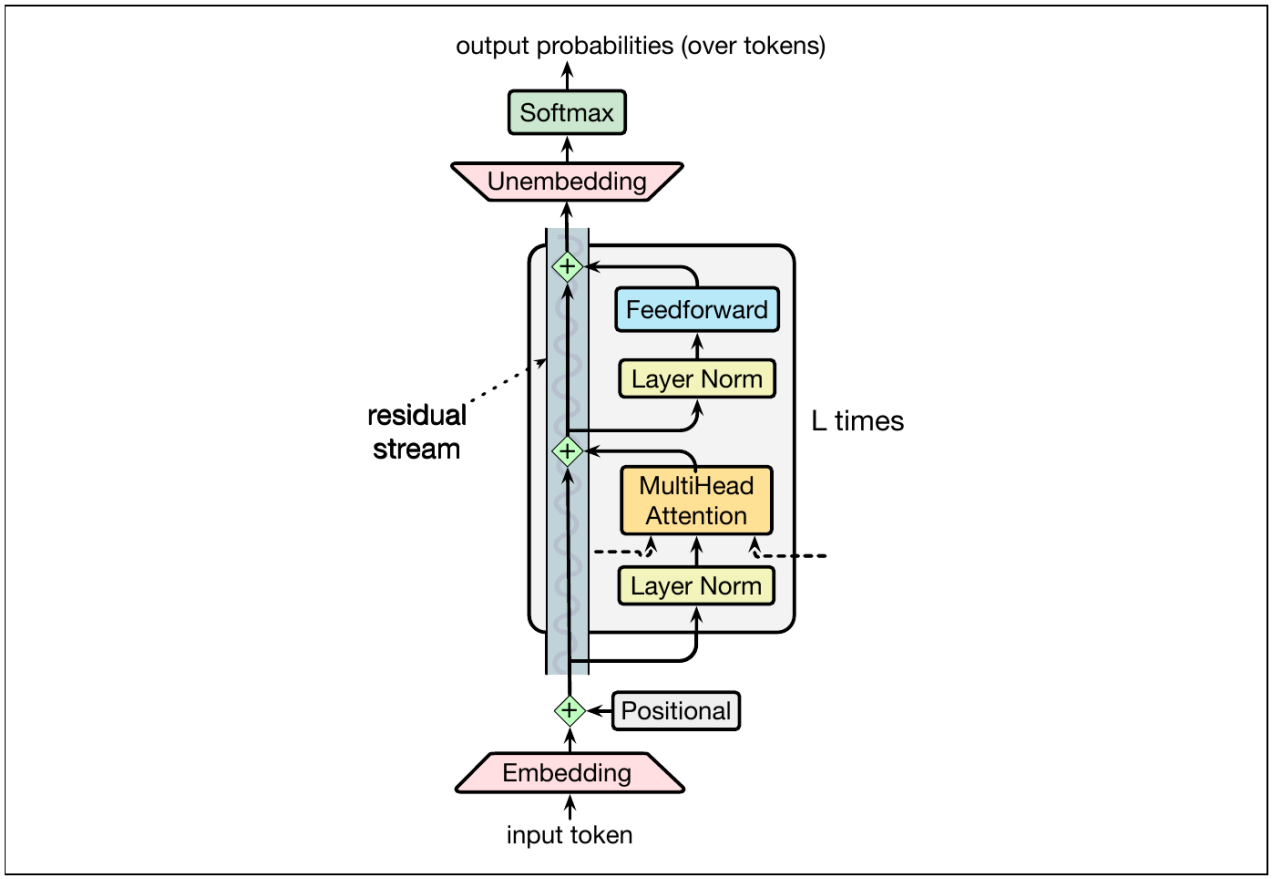
The transformer



Vaswani et al. 2017

Attention and the transformer architecture

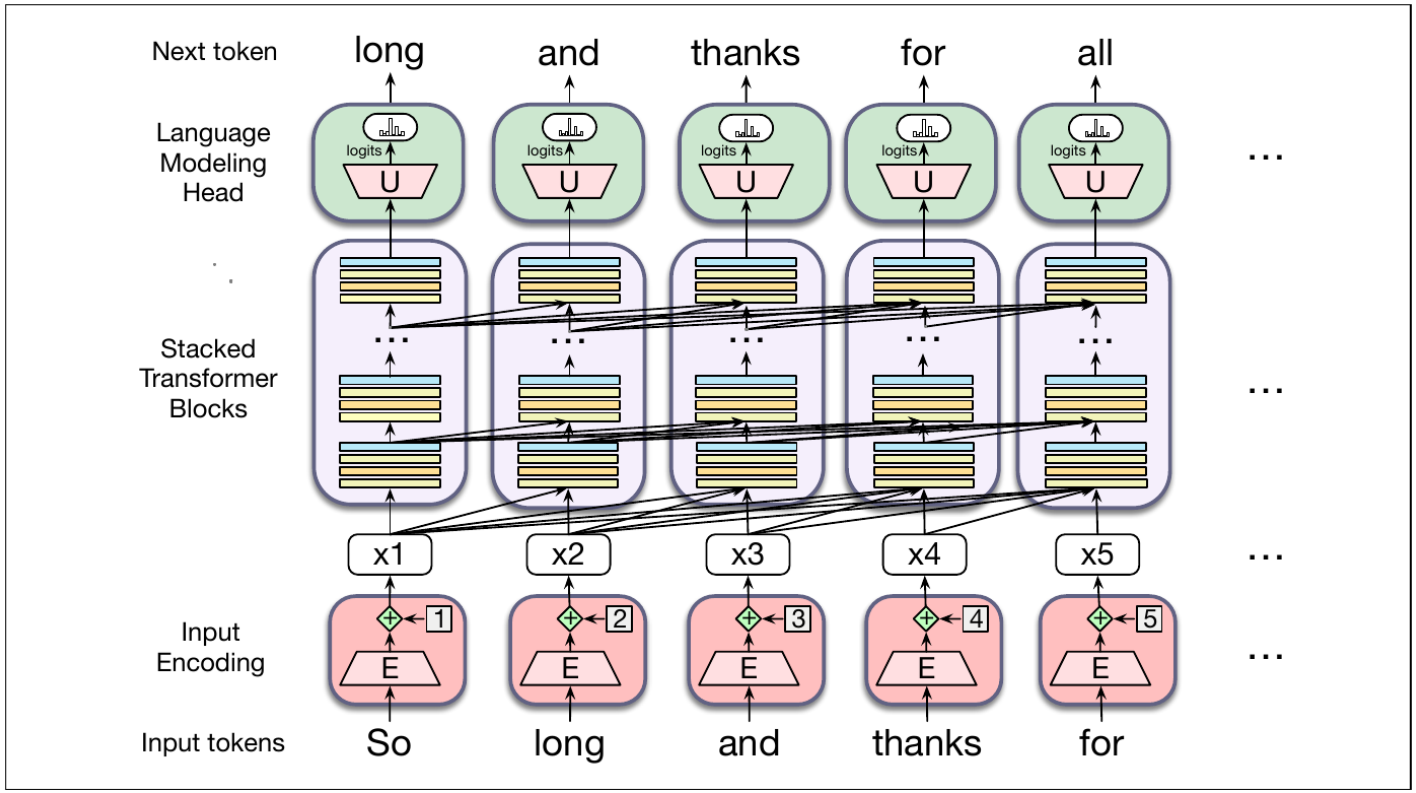
Inside the decoder



The residual stream carries a token's representation through transformer blocks, with attention adding information from other tokens. Jurafsky and Martin 2026

Attention and the transformer architecture

Stacked transformer blocks

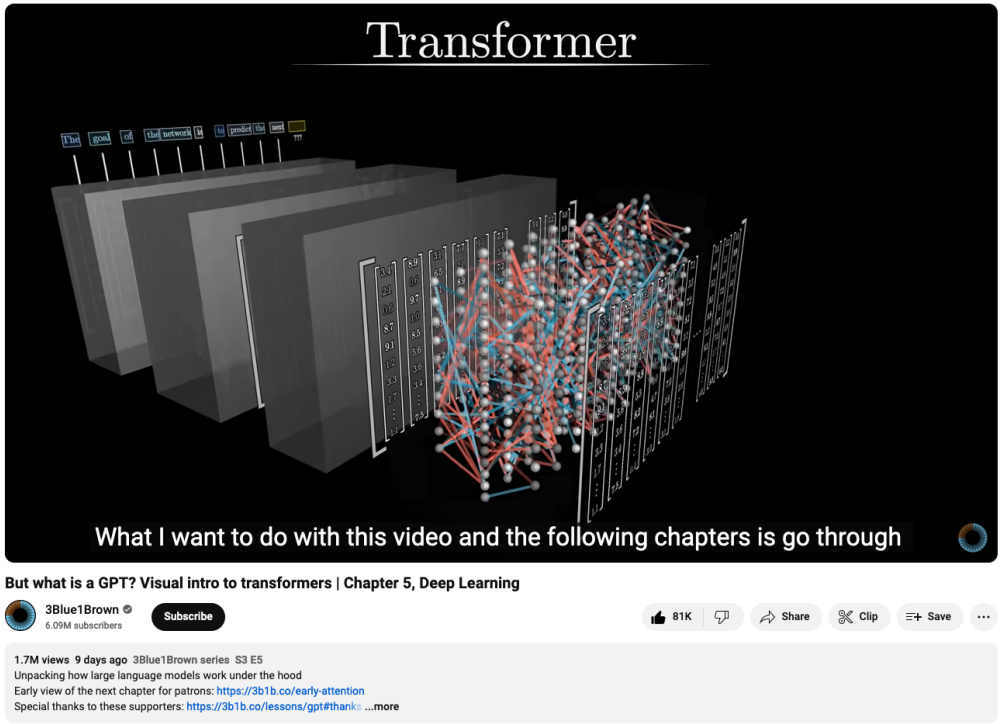


Each input token is encoded, passed through stacked transformer blocks, and used to predict the next token.
Jurafsky and Martin 2026

Attention and the transformer architecture

Further resources

- 3Blue1Brown has a series on YouTube explaining how attention and transformers work



Part II: Large language models

Large language models

BERT

BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding

Jacob Devlin **Ming-Wei Chang** **Kenton Lee** **Kristina Toutanova**
Google AI Language
{jacobdevlin,mingweichang,kentonl,kristout}@google.com

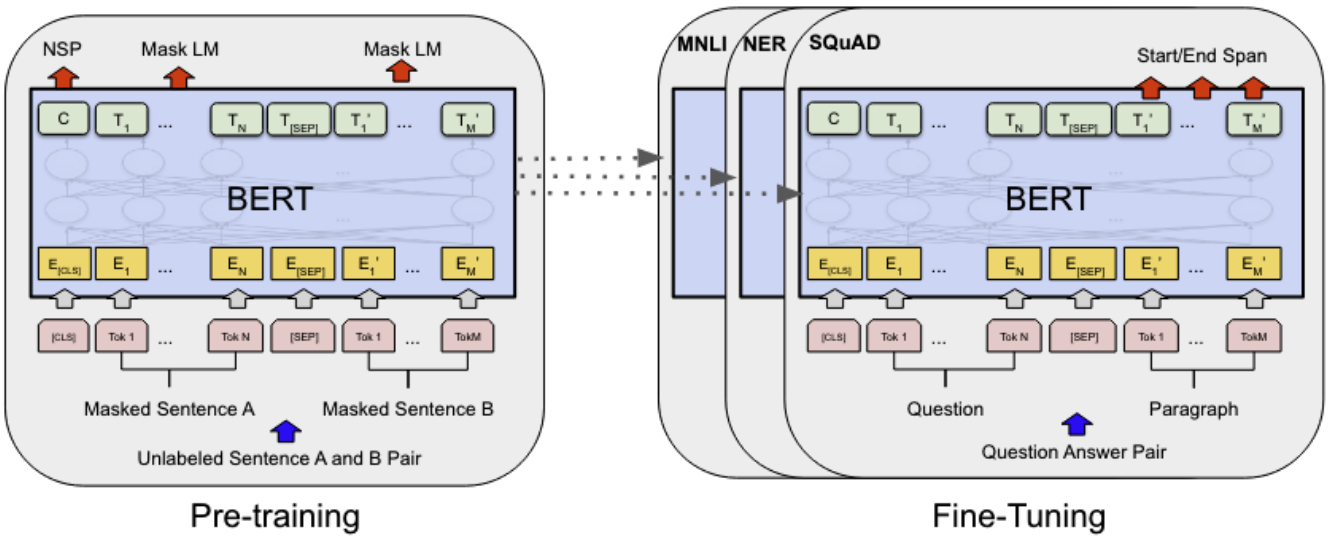
Abstract

We introduce a new language representation model called **BERT**, which stands for **B**idirectional **E**ncoder **R**epresentations from **T**ransformers. Unlike recent language representation models (Peters et al., 2018a; Radford et al., 2018), BERT is designed to pre-train deep bidirectional representations from unlabeled text by jointly conditioning on both left and right context in all layers. As a result, the pre-trained BERT model can be fine-tuned with just one additional output layer to create state-of-the-art models for a wide range of tasks, such as question answering and language inference, without substantial task-specific architecture modifications.

There are two existing strategies for applying pre-trained language representations to downstream tasks: *feature-based* and *fine-tuning*. The feature-based approach, such as ELMo (Peters et al., 2018a), uses task-specific architectures that include the pre-trained representations as additional features. The fine-tuning approach, such as the Generative Pre-trained Transformer (OpenAI GPT) (Radford et al., 2018), introduces minimal task-specific parameters, and is trained on the downstream tasks by simply fine-tuning *all* pre-trained parameters. The two approaches share the same objective function during pre-training, where they use unidirectional language models to learn general language representations.

Large language models

BERT: pre-training and fine-tuning



Devlin et al. 2019

Large language models

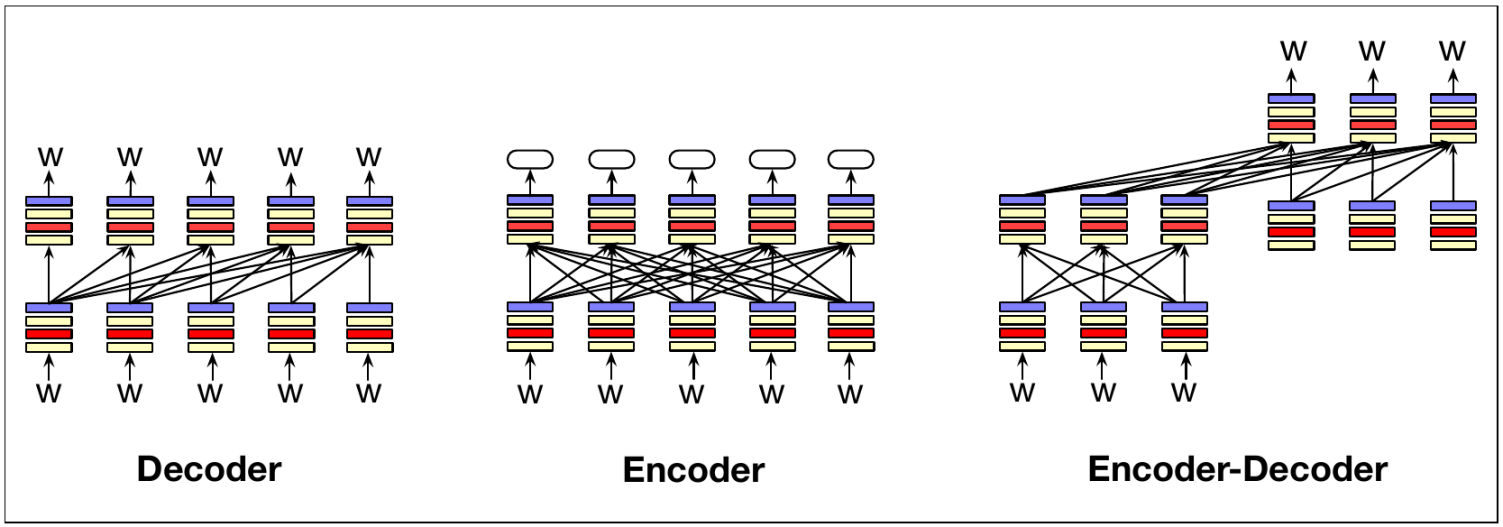
Foundation models and transfer learning

- Large language models (LLMs) are described as **foundation models** insofar as pre-trained models serve as a foundation for other tasks (Bommasani et al. 2021)
- **Transfer learning** describes the process through which a model is adapted to new tasks

Bommasani, Rishi, et al. 2021. "On the Opportunities and Risks of Foundation Models." *arXiv*:2108.07258.

Large language models

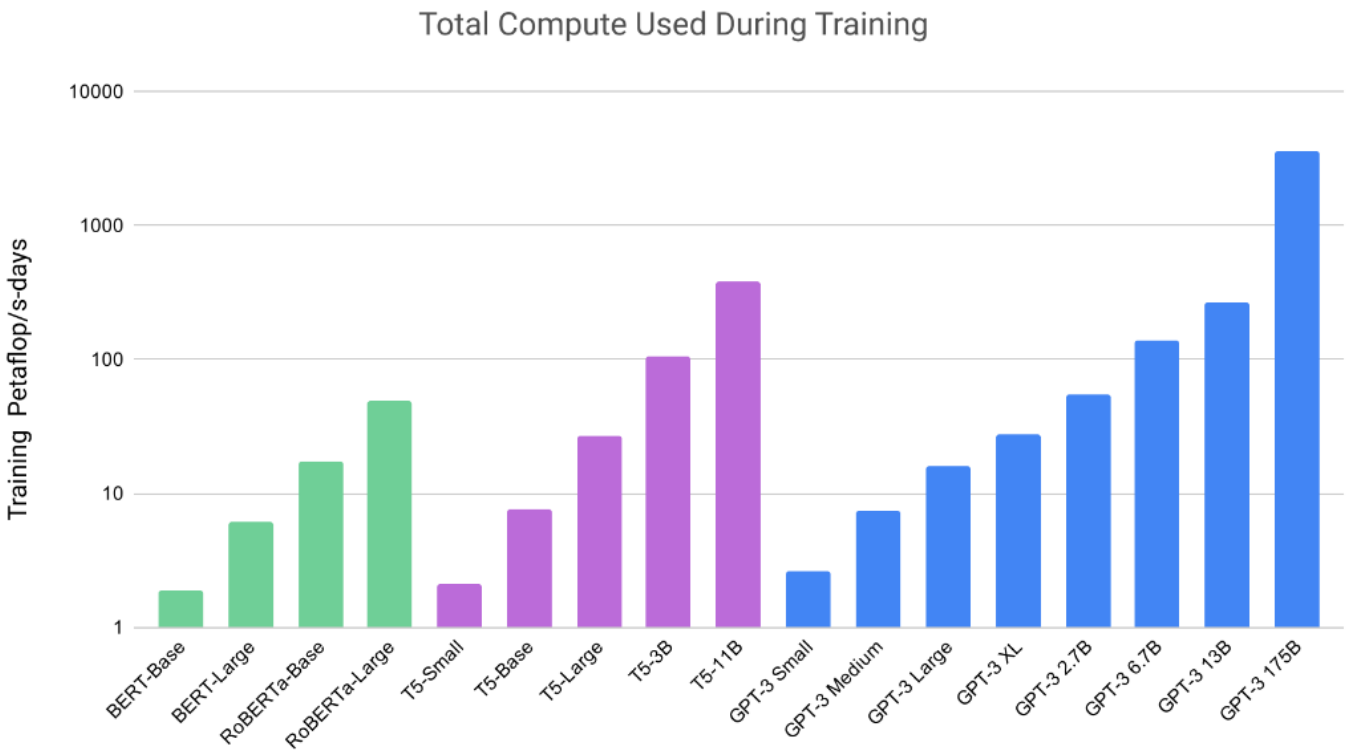
Decoders, encoders, and encoder-decoders



Decoders take tokens as input and generate tokens. Encoders produce a vector representation of each input token. Encoder-decoders take tokens as input and generate a series of tokens. Jurafsky and Martin 2026

Large language models

From BERT to GPT-3

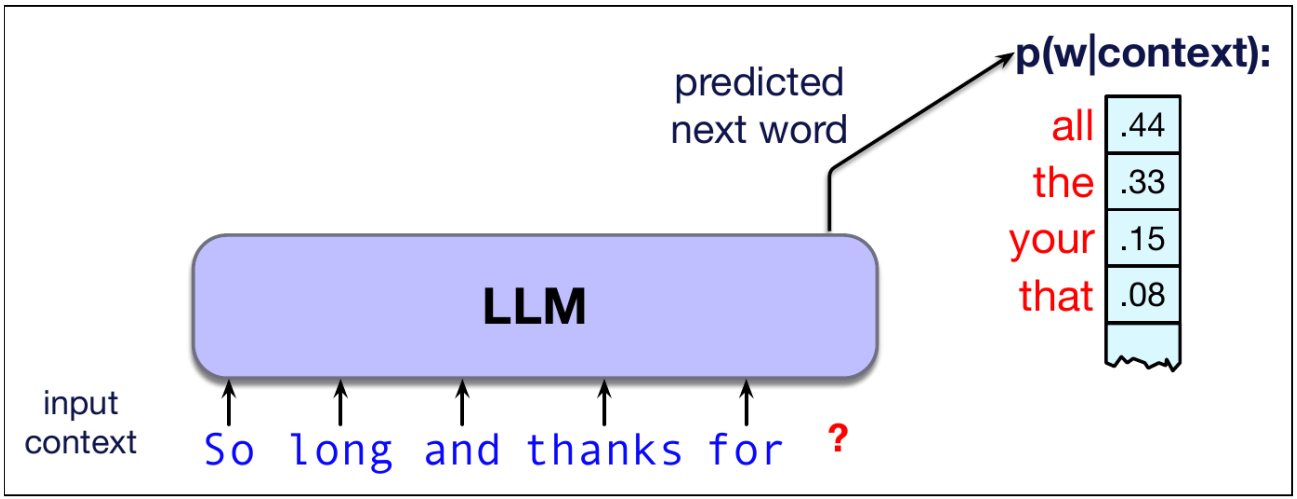


Brown et al. 2020

Generating text

Predicting the next word

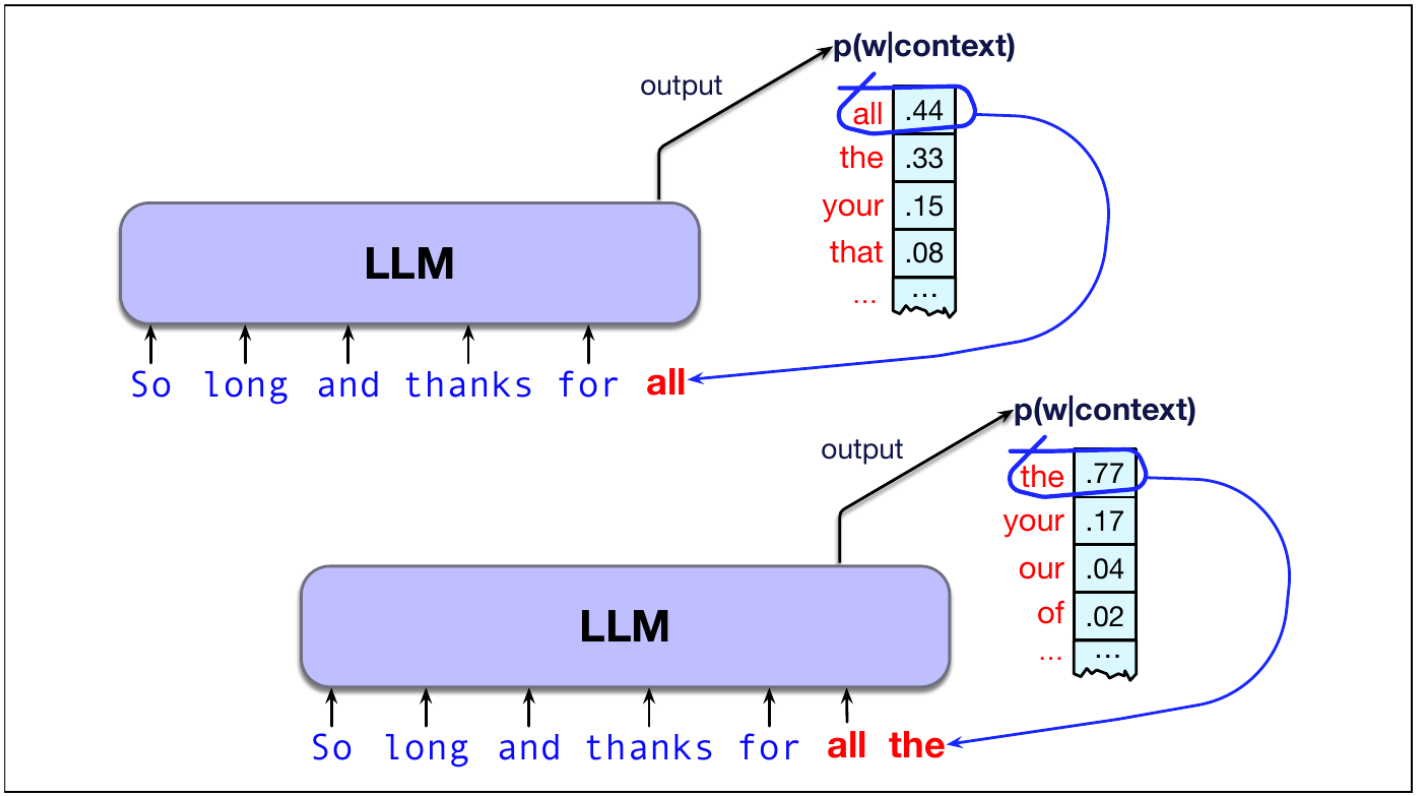
LLMs predict the next *token*, usually a word or part of a word.



An LLM takes a context as input and outputs a probability distribution over possible next words. Jurafsky and Martin 2026

Generating text

Autoregressive generation



The model generates text by sampling a word from the distribution, adding it to the context, and repeating the process. Jurafsky and Martin 2026

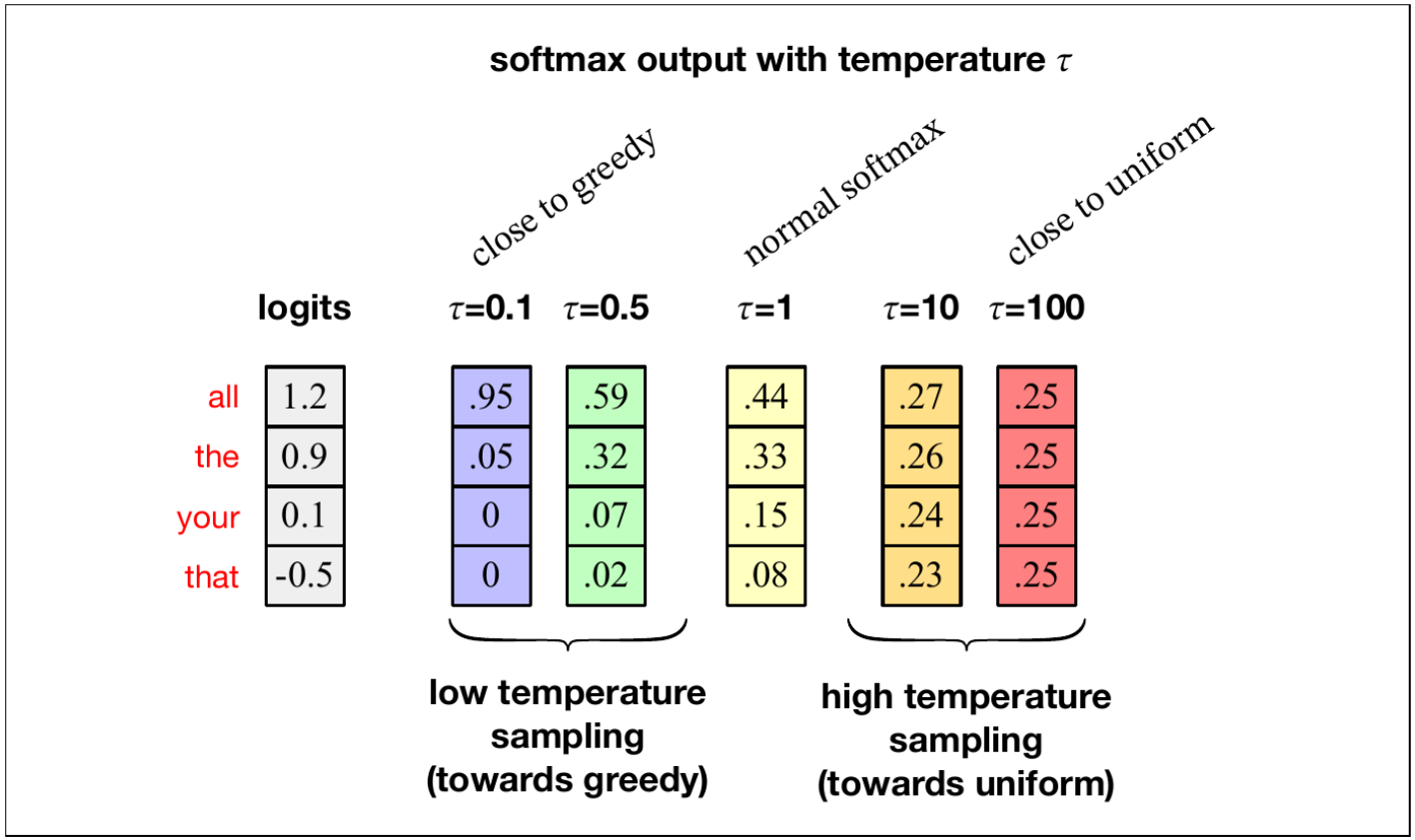
Generating text

Sampling and temperature

- The *temperature* controls how random the sampling is
  - Near 0, the model always picks the most likely word, making it more deterministic
  - Higher temperatures make it more likely that less probable words are used

Generating text

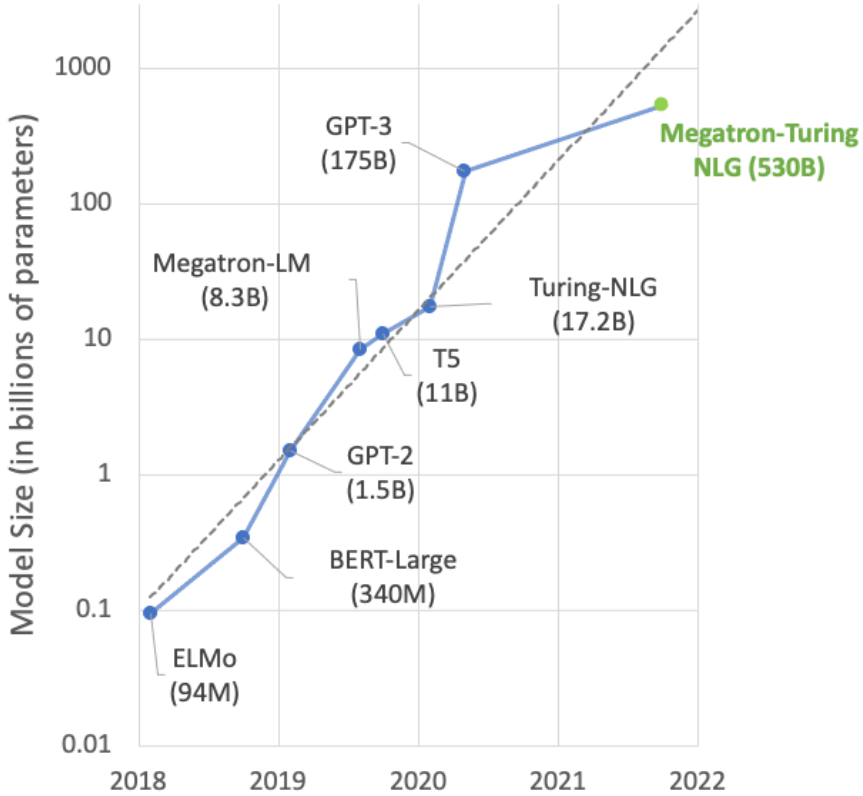
Sampling and temperature



At a temperature near 0, sampling becomes "greedy" decoding. Jurafsky and Martin 2026

Scale

Model size over time

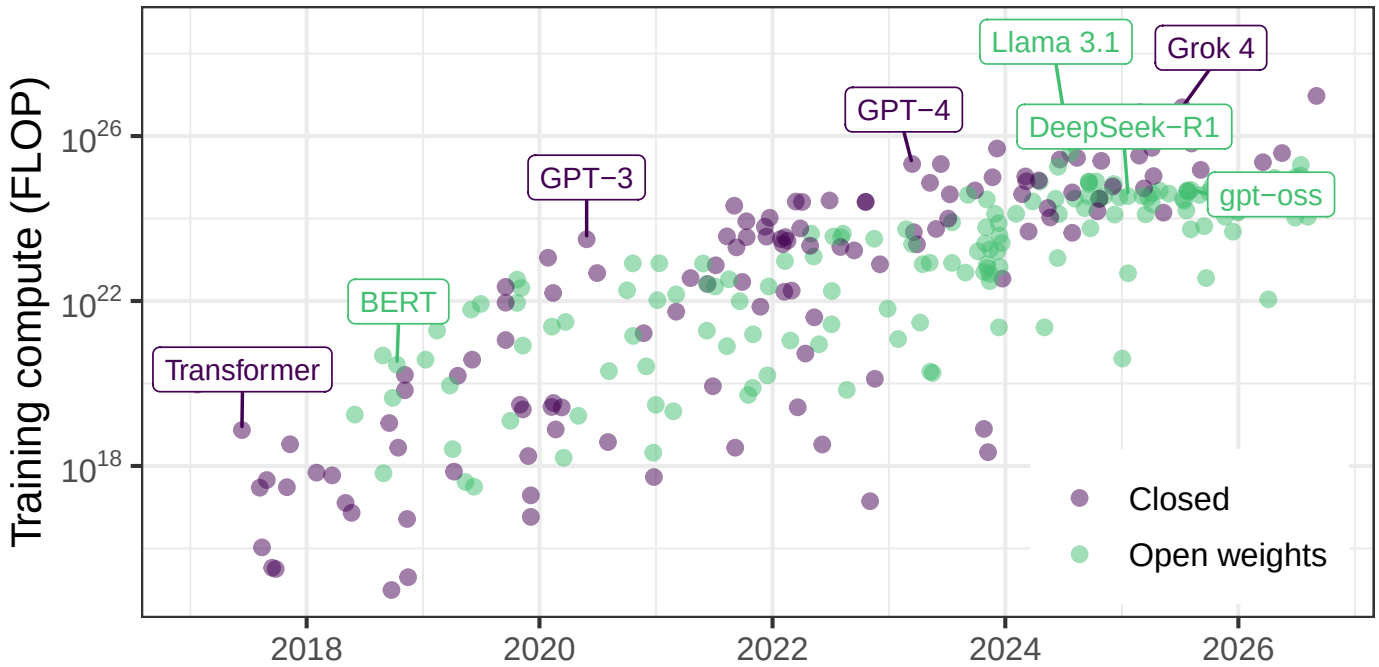


https://developer.nvidia.com/blog/using-deepspeed-and-megatron-to-train-megatron-turing-nlg-530b-the-worlds-largest-and-most-powerful-generative-language-model/

Scale

Training compute over time

Most companies no longer disclose the size of their models, so the compute used for training is a common measure of scale.

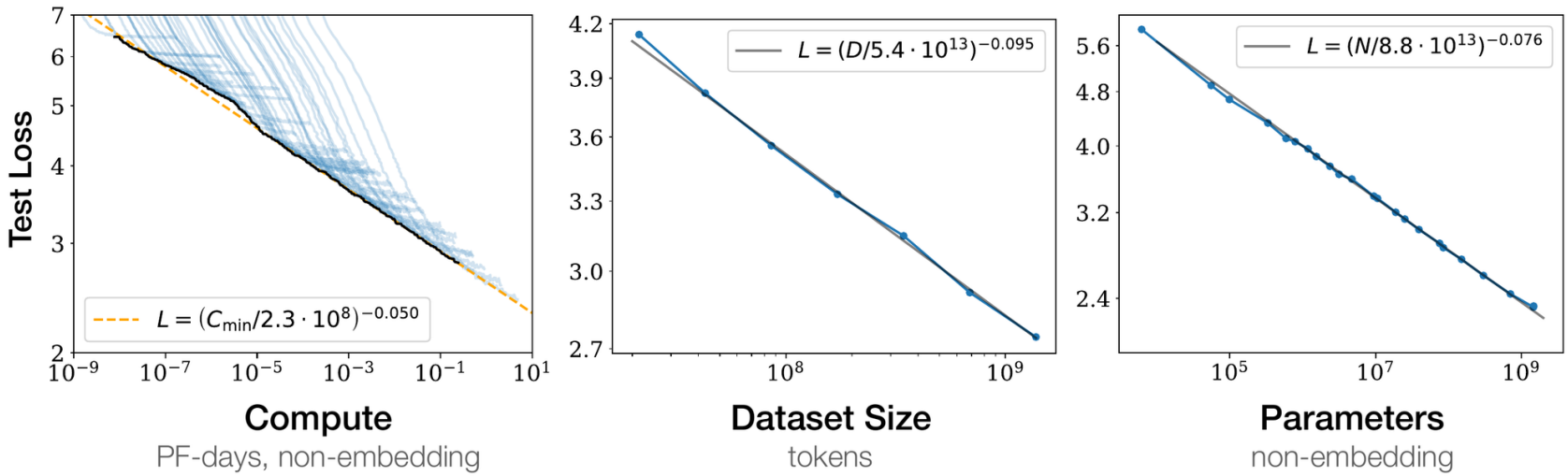


Data from Epoch AI. 2026. "Data on AI Models." Accessed October 22, 2026. https://epoch.ai/data/ai-models.

Scale

Scaling laws

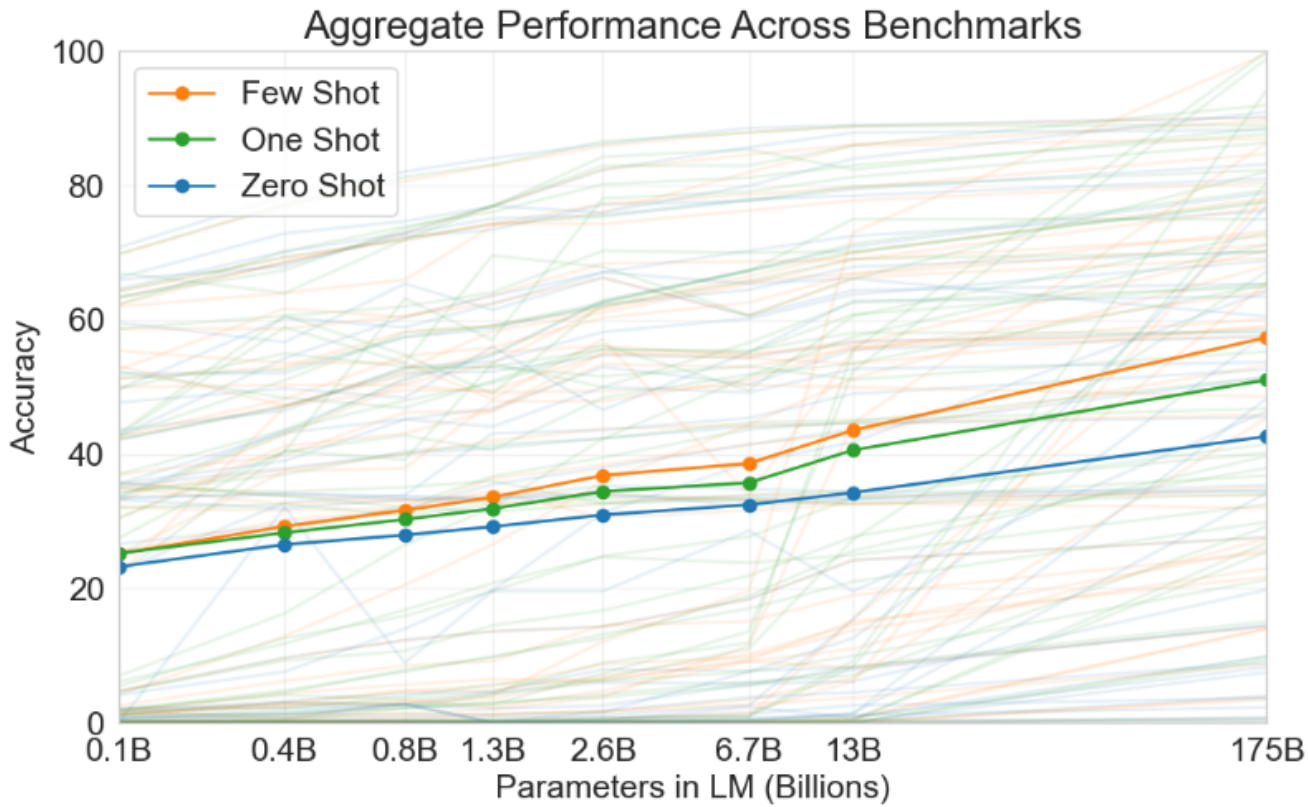
Language model performance improves as compute, data, and parameters increase



Kaplan, Jared, Sam McCandlish, Tom Henighan, Tom B. Brown, Benjamin Chess, Rewon Child, Scott Gray, Alec Radford, Jeffrey Wu, and Dario Amodei. 2020. "Scaling Laws for Neural Language Models." *arXiv*:2001.08361.

Scale

Scale and performance



Brown et al. 2020

Scale

Vast training data

Dataset	Quantity (tokens)	Weight in training mix	Epochs elapsed when training for 300B tokens
Common Crawl (filtered)	410 billion	60%	0.44
WebText2	19 billion	22%	2.9
Books1	12 billion	8%	1.9
Books2	55 billion	8%	0.43
Wikipedia	3 billion	3%	3.4

Brown et al. 2020

Prompting

Text-to-text models

- Early models like BERT work more like standard machine learning techniques
  - e.g. extract an embedding for a term, fine-tune to predict a label given the input
- Text-to-text models take text as input and produce text as output*
- This natural language interface makes it easy to transfer to new tasks and enables chat interfaces

* Raffel, Colin, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J. Liu. 2020. "Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer." *Journal of Machine Learning Research* 21(140): 1–67.

Prompting

Context windows

- The *context window* defines the amount of text a model can process at once, measured in tokens
- Context windows have been increasing over time:
  - BERT (2018): 512
  - GPT-3 (2020): 2,048
  - GPT-4 Turbo (2023): 128,000
  - Claude 3 (2024): 200,000
  - Gemini 1.5 Pro (2024): 1,000,000

Prompting

User and system prompts

- *Prompts* guide the model to generate specific outputs
  - **User prompts** result in a single output
    - e.g. "Is the following text positive or negative?"
  - **System prompts** guide overall model behavior
    - e.g. "You are a helpful assistant . . . "

Prompting

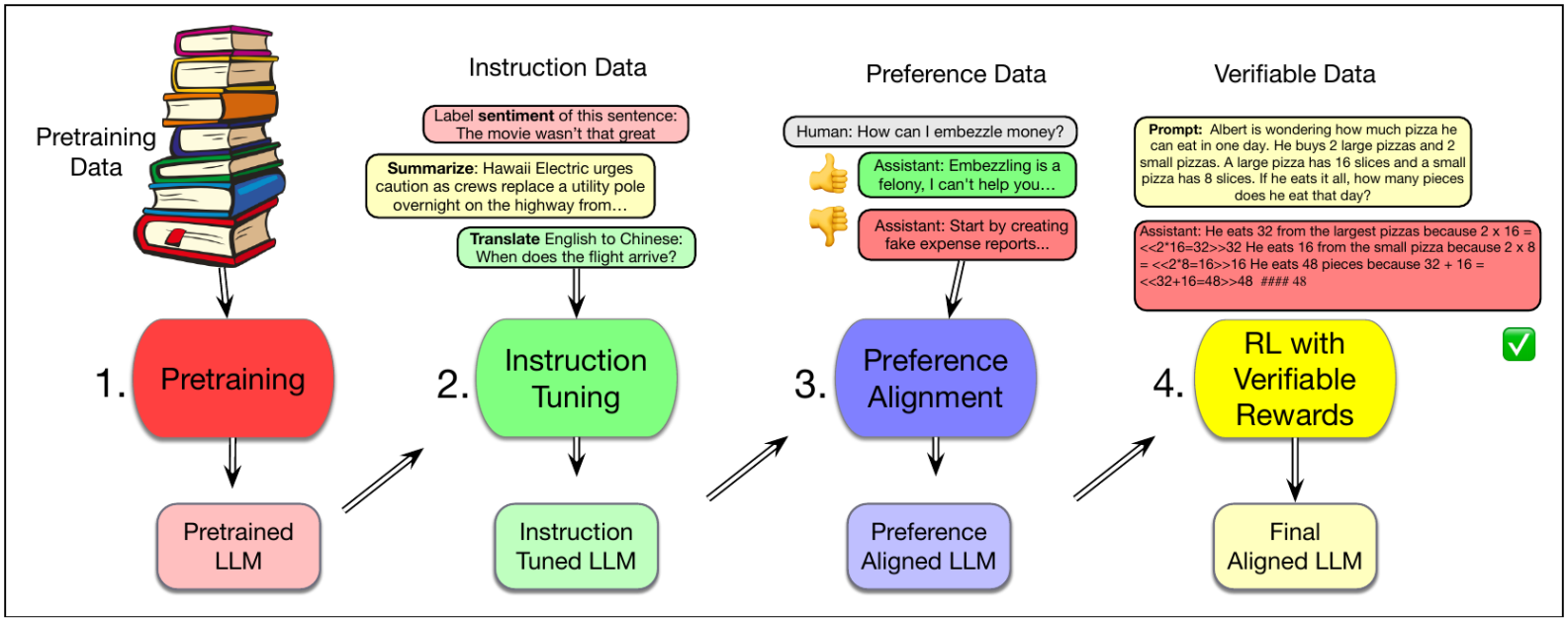
Prompt engineering

- An emerging field known as *prompt engineering* considers how to effectively design prompts
  - Art or science?
- Some systems can automatically refine and enhance prompts
  - e.g. ChatGPT rewrote user prompts to DALL-E 3 to make better images, and to diversify demographic representation in outputs

Part III: Post-training, reasoning, and agents

Post-training

Four stages of training



Jurafsky and Martin 2026

Post-training

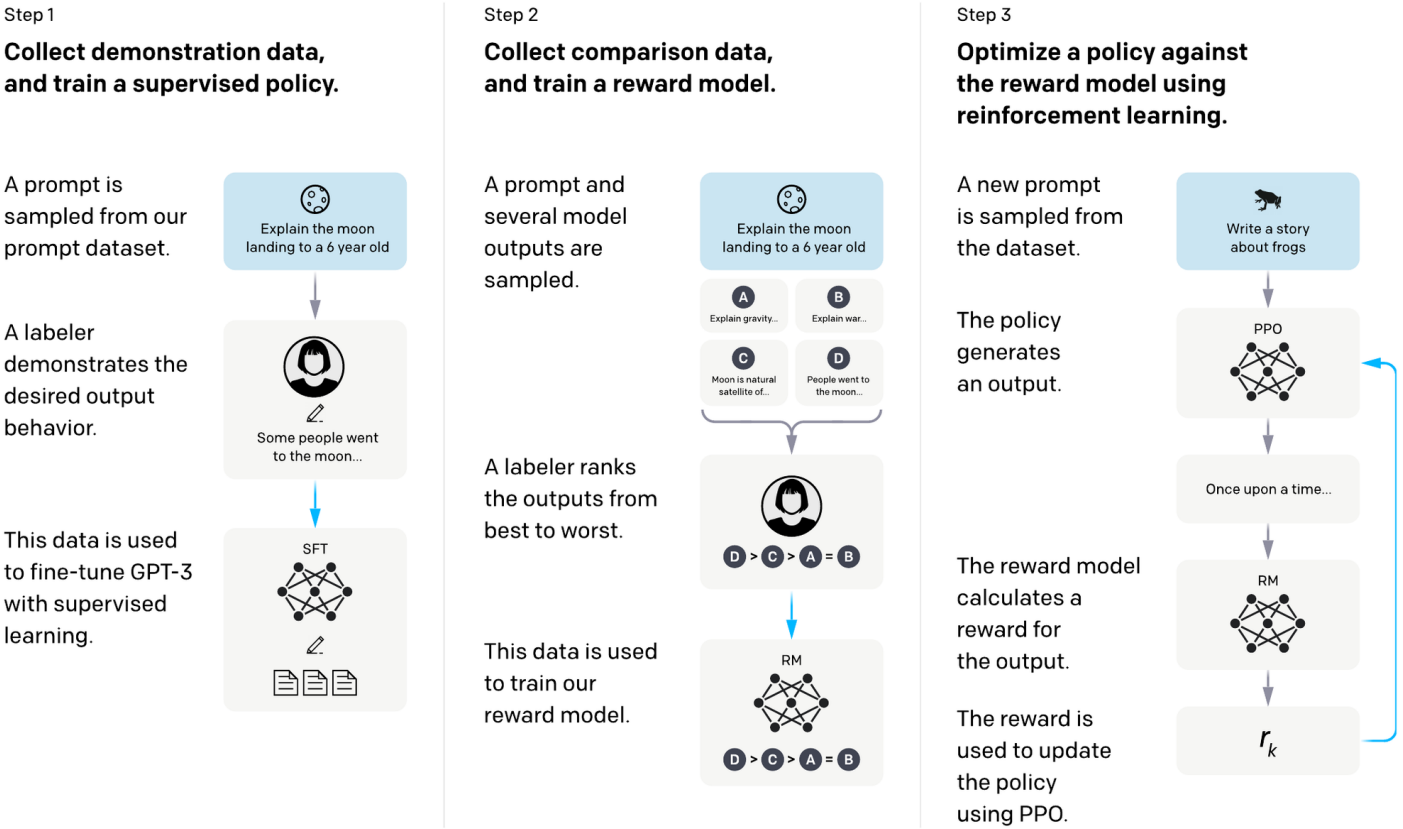
Reinforcement learning with human feedback (RLHF)

- LLMs are pre-trained as language models but can be adapted to more general tasks by further training
- RLHF involves using feedback to guide a model's behavior and is critical to contemporary chatbots
  - Feedback used to make models "more helpful" and "less harmful"

Ouyang, Long, et al. 2022. "Training Language Models to Follow Instructions with Human Feedback." In *Advances in Neural Information Processing Systems* 35, 27730–44.

Post-training

RLHF phases



Ouyang et al. 2022

Post-training

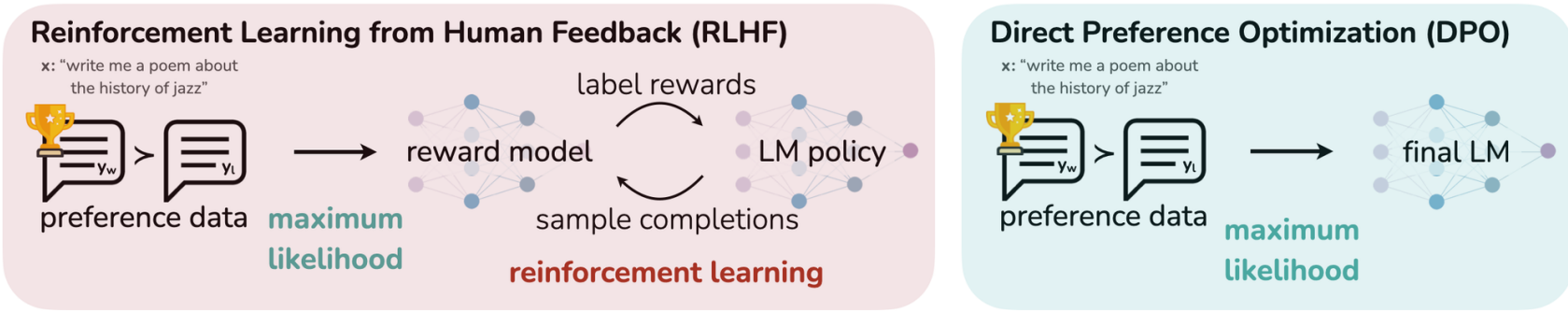
Direct preference optimization (DPO)

- RLHF trains a separate *reward model* on human preferences, then uses reinforcement learning to optimize the LLM against it
- DPO skips the reward model and trains the LLM directly on pairs of preferred and rejected responses
  - Simpler, more stable, and cheaper than RLHF

Rafailov, Rafael, Archit Sharma, Eric Mitchell, Christopher D. Manning, Stefano Ermon, and Chelsea Finn. 2023. "Direct Preference Optimization: Your Language Model Is Secretly a Reward Model." In *Advances in Neural Information Processing Systems* 36, 53728–41.

Post-training

RLHF versus DPO

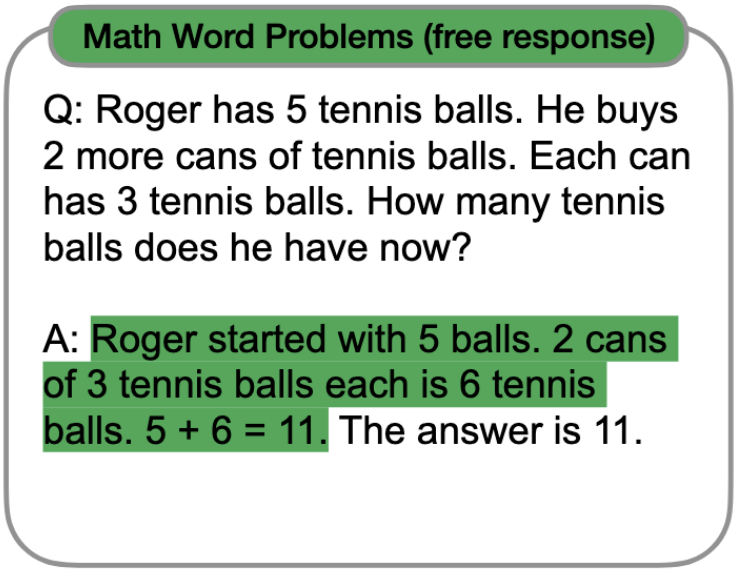


Rafailov et al. 2023

Reasoning models

Chain-of-thought prompting

- Including examples of step-by-step reasoning in the prompt improves performance on math and reasoning problems



Wei, Jason, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed H. Chi, Quoc V. Le, and Denny Zhou. 2022. "Chain-of-Thought Prompting Elicits Reasoning in Large Language Models." In *Advances in Neural Information Processing Systems* 35, 24824–37.

Reasoning models

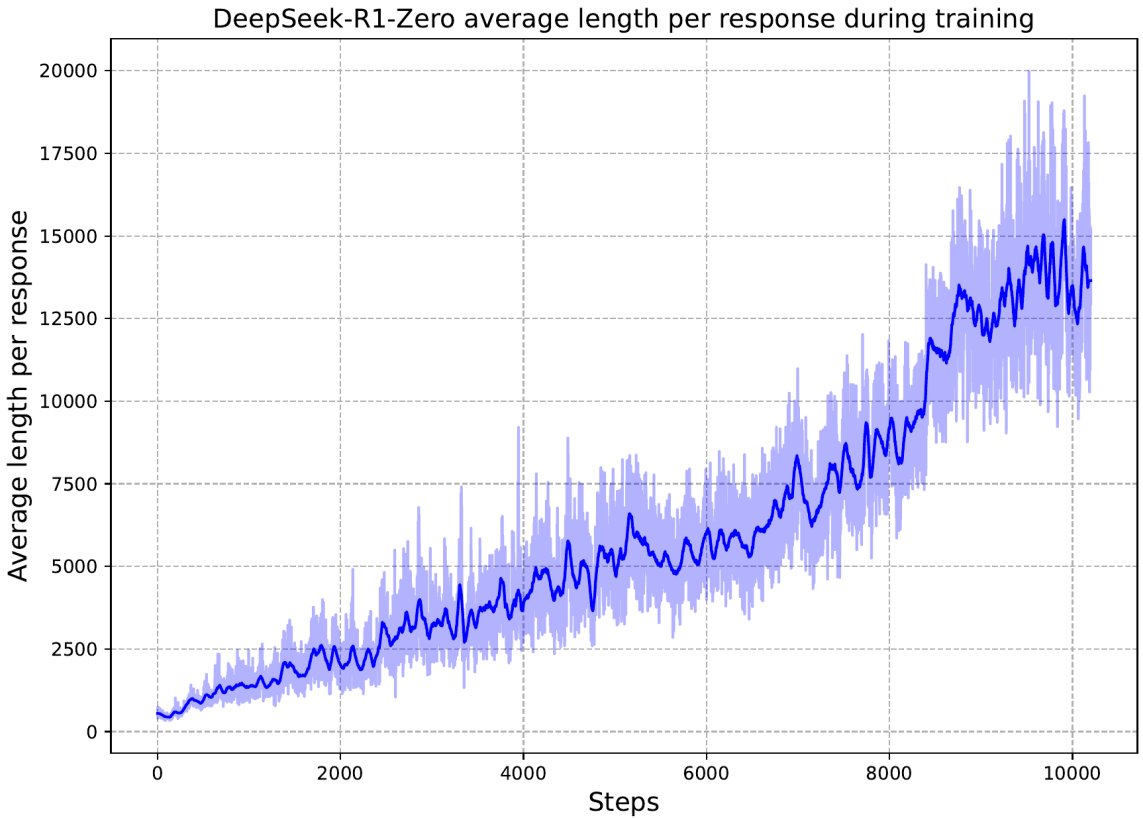
Learning to reason

- Recent models are further trained using reinforcement learning to generate intermediate reasoning steps before answering
  - e.g. OpenAI's o1, DeepSeek-R1
- Rewards come from tasks with verifiable answers, such as math problems and coding tests

Guo, Daya, et al. 2025. "DeepSeek-R1 Incentivizes Reasoning in LLMs Through Reinforcement Learning." *Nature* 645(8081): 633–38.

Reasoning models

"Thinking" longer improves performance



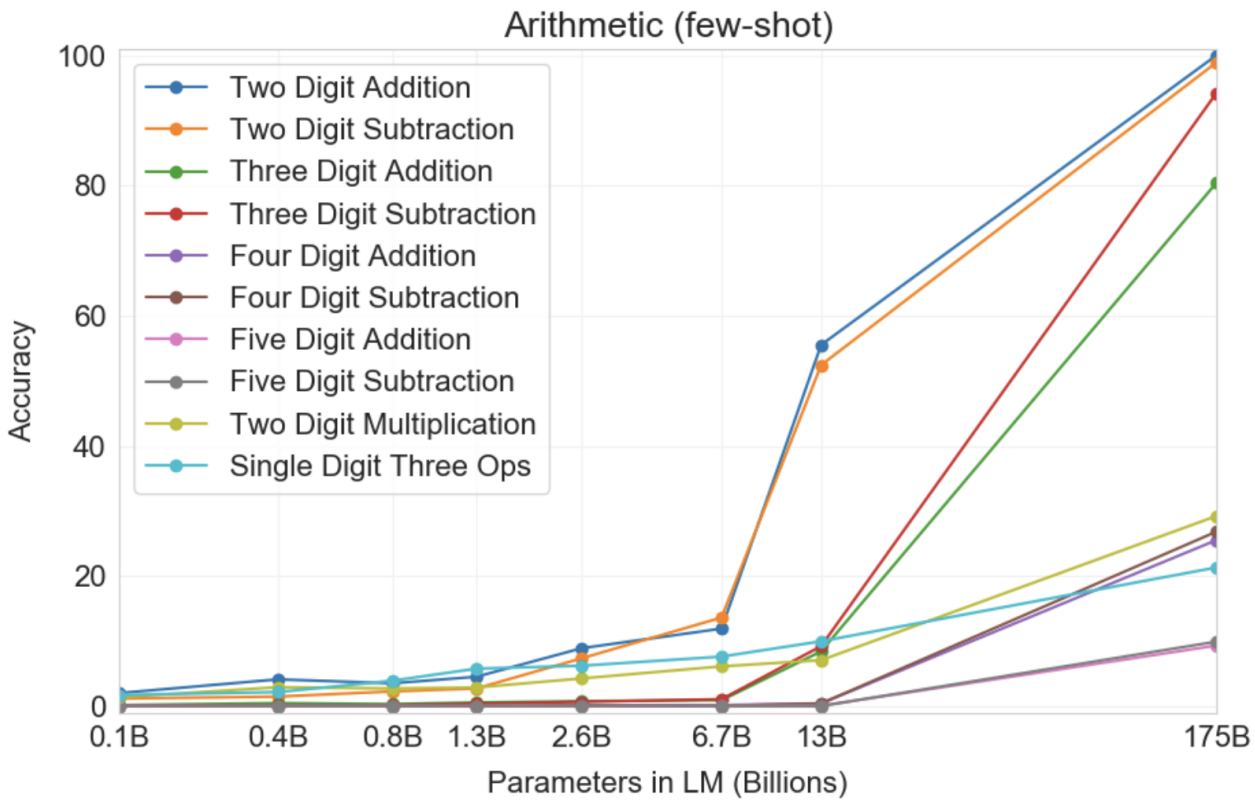
Guo et al. 2025

Agents

What are agents?

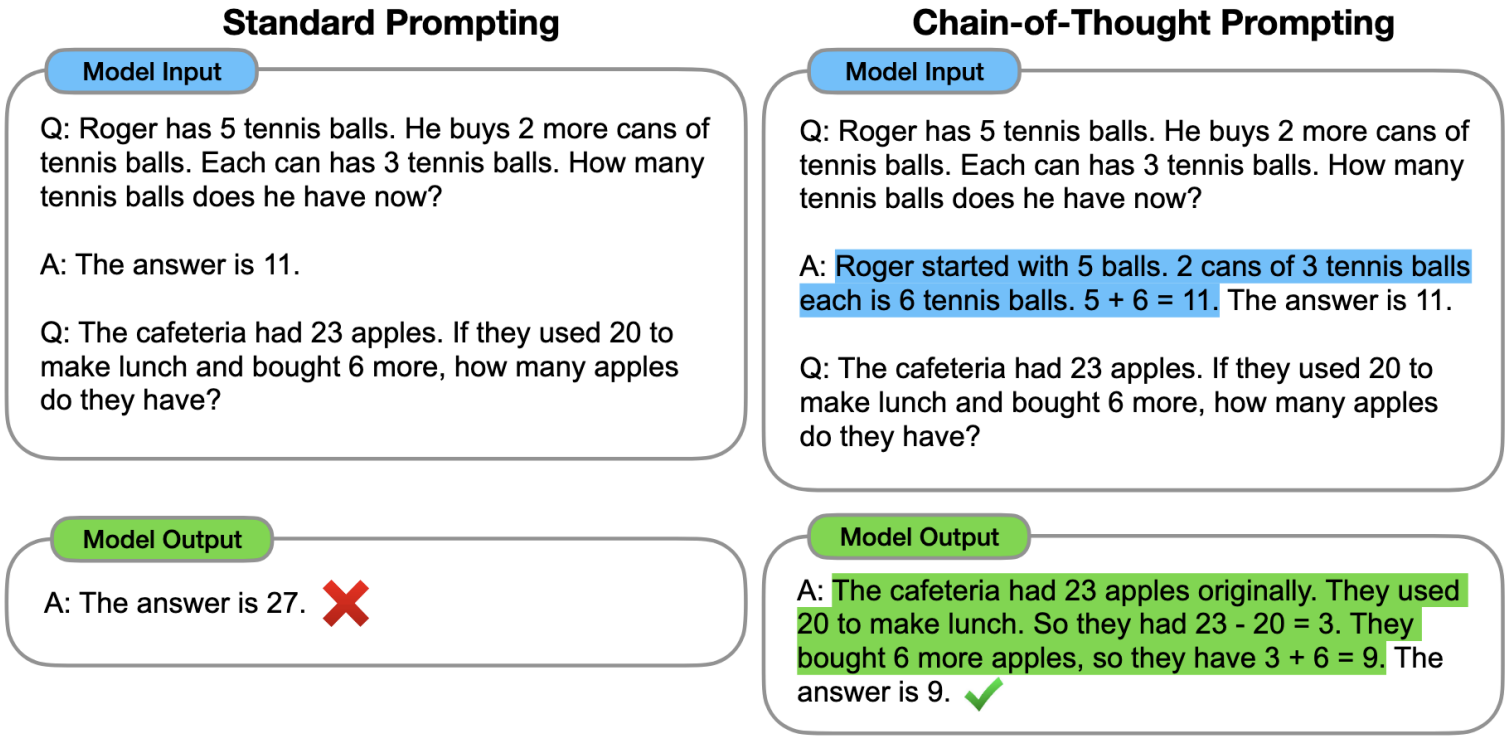
- LLMs are increasingly used as *agents* that can use tools and pursue a goal over many steps
  - The model chooses an action, such as searching the web or running code, observes the result, and decides what to do next
- Coding agents like Claude Code and OpenAI's Codex can write, run, and debug code across a project

Advancing capabilities, 2020–2026



GPT-3's few-shot accuracy on arithmetic problems. Brown et al. 2020

Advancing capabilities, 2020–2026



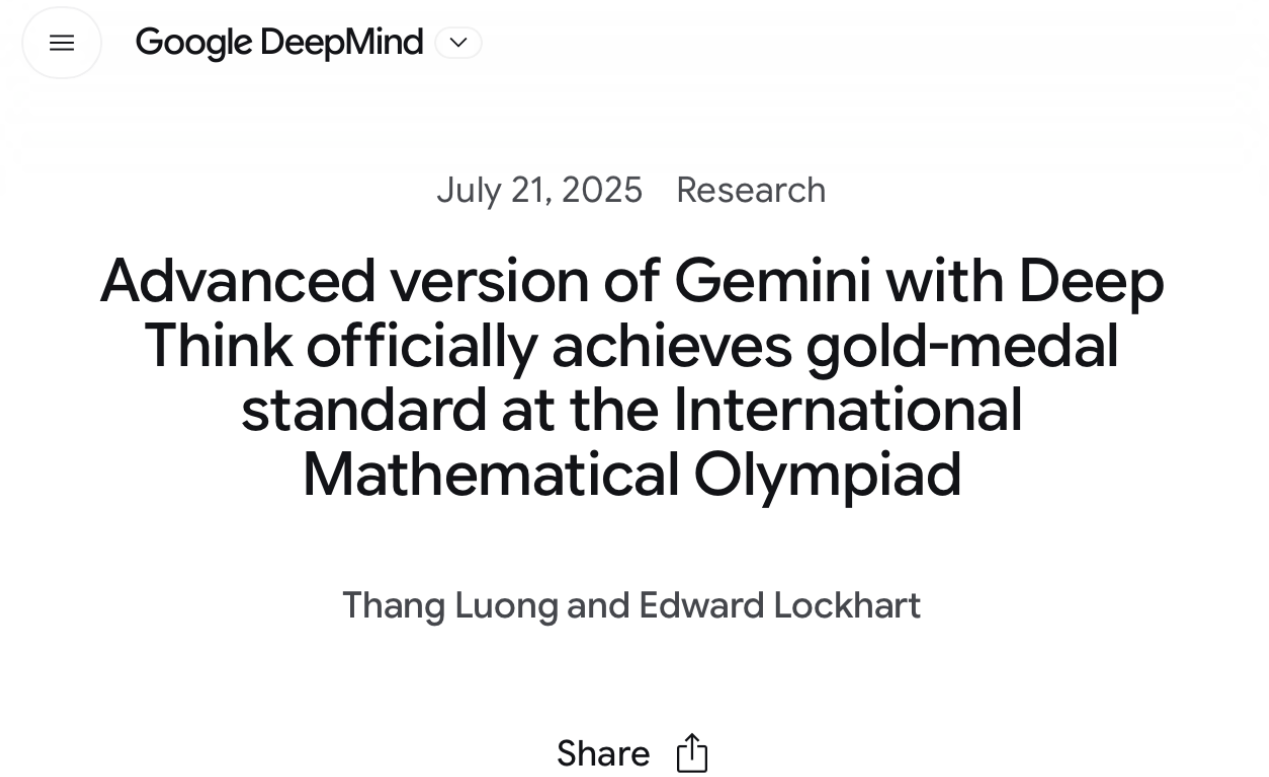
Wei et al. 2022

Advancing capabilities, 2020–2026

> *I have played a little bit with OpenAI's new iteration of #GPT, GPT-o1, which performs an initial reasoning step before running the LLM. . . The experience seemed roughly on par with trying to advise a mediocre, but not completely incompetent, (static simulation of a) graduate student.*
> — *Terence Tao, September 2024*

https://mathstodon.xyz/@tao/113132502735585408

Advancing capabilities, 2020–2026



Google DeepMind

July 21, 2025 Research

Advanced version of Gemini with Deep Think officially achieves gold-medal standard at the International Mathematical Olympiad

Thang Luong and Edward Lockhart

Share

https://deepmind.google/discover/blog/advanced-version-of-gemini-with-deep-think-officially-achieves-gold-medal-standard-at-the-international-mathematical-olympiad/

Advancing capabilities, 2020–2026

A Millennium Prize Problem

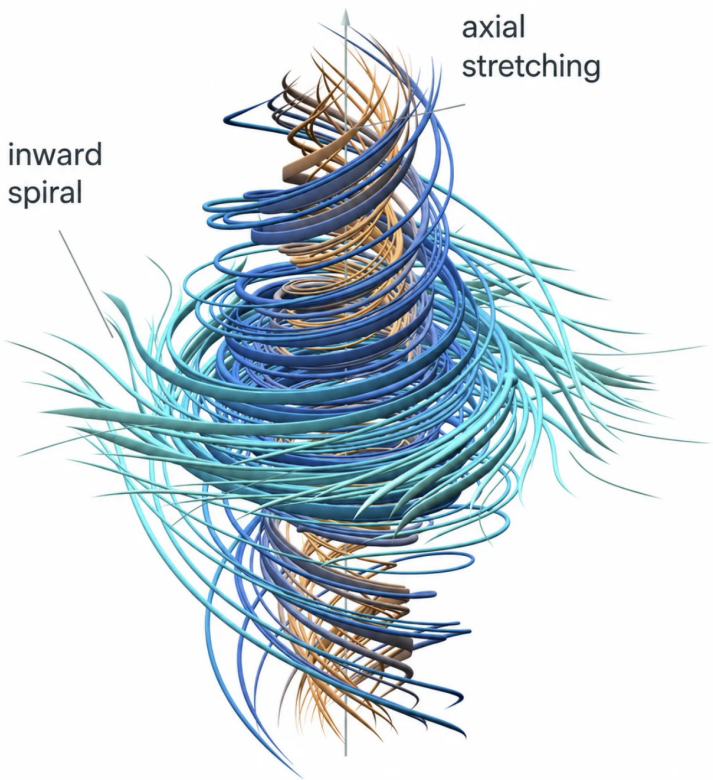
OpenAI Says It Has Cracked One of Math's 'Millennium Problems'

The company's announcement is the most dramatic sign yet that artificial intelligence is fundamentally transforming the field of higher mathematics.

https://www.nytimes.com/2026/09/08/science/openai-proof-millennium-problem.html

Advancing capabilities, 2020–2026

The Navier–Stokes equations



OpenAI. 2026. "On the Navier–Stokes Millennium Prize Problem." September 8.

Advancing capabilities, 2020–2026

We used a system of coordinating agents powered by our internal model. The agents had access to tools such as the ability to read from a cached version of the internet and the ability to run code. Agents were subdivided into groups with the ability to communicate within the group. The groups varied in size, and the group that produced the Navier–Stokes resolution involved on the order of 10,000 concurrent agents. At all times we maintained the same strict safeguards that we apply to all our frontier model evaluations, including monitoring and isolation.

OpenAI 2026

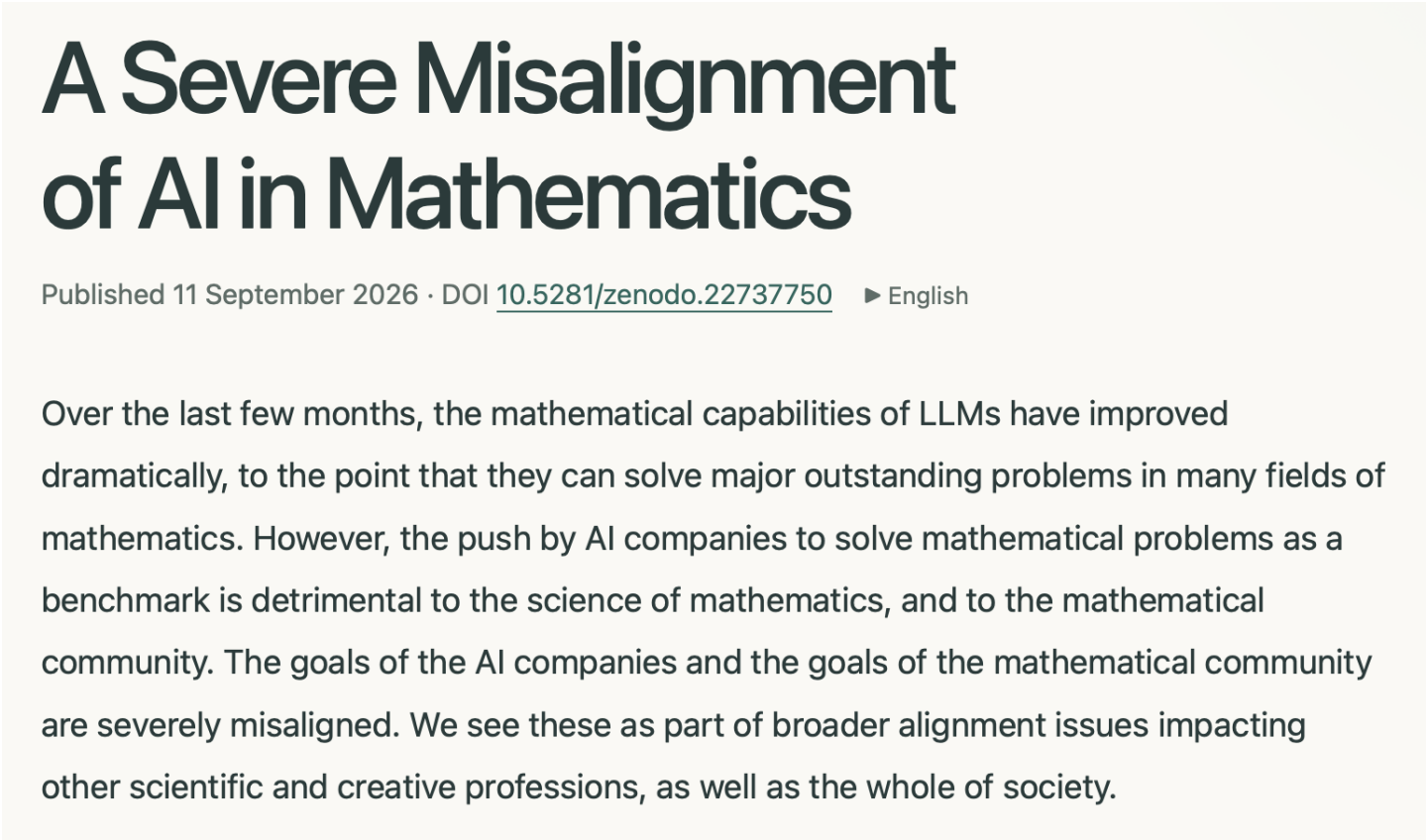
Advancing capabilities, 2020–2026

The agents arrived at their resolution on Saturday, September 5, about 88 hours after the first agents were launched. Lean formalization and verification took an additional 17 hours via GPT-6 Astra.

Across all attempted problems, the agents sent 4.9 million messages and used about 300 billion output tokens. In the process of resolving the Navier–Stokes problem, the agents sent 2.7 million messages and used approximately 130 billion output tokens.

OpenAI 2026

Advancing capabilities, 2020–2026



A Severe Misalignment of AI in Mathematics

Published 11 September 2026 · DOI 10.5281/zenodo.22737750 ▶ English

Over the last few months, the mathematical capabilities of LLMs have improved dramatically, to the point that they can solve major outstanding problems in many fields of mathematics. However, the push by AI companies to solve mathematical problems as a benchmark is detrimental to the science of mathematics, and to the mathematical community. The goals of the AI companies and the goals of the mathematical community are severely misaligned. We see these as part of broader alignment issues impacting other scientific and creative professions, as well as the whole of society.

Avila, Artur, et al. 2026. "A Severe Misalignment of AI in Mathematics." Zenodo. https://doi.org/10.5281/zenodo.22737750

Advancing capabilities, 2020–2026

OpenAI Releases Findings on 377 Math Problems, Further Roiling Field

After a previous solution angered mathematicians, the company characterized the new release as being more responsive to concerns about A.I. disrupting research.

https://www.nytimes.com/2026/10/06/science/openai-math-problems.html

Benchmarks

Humanity's Last Exam

- 2,500 questions written by nearly 1,000 experts in over 100 subjects, included only if leading LLMs could not answer them
- Best score at launch in January 2025 was below 10%

Center for AI Safety, Scale AI, and HLE Contributors Consortium. 2026. "A Benchmark of Expert-Level Academic Questions to Assess AI Capabilities." *Nature* 649(8099): 1139–46.

Benchmarks

Humanity's Last Exam

Classics

Question:



Here is a representation of a Roman inscription, orginally found on a tombstone. Provide a translation for the Palmyrene script.
A transliteration of the text is provided: RGYNᵓ BT HRY BR ᶜTᵓ HBL

Henry T
Merton College, Oxford

Ecology

Question:

Hummingbirds within Apodiformes uniquely have a bilaterally paired oval bone, a sesamoid embedded in the caudolateral portion of the expanded, cruciate aponeurosis of insertion of m. depressor caudae. How many paired tendons are supported by this sesamoid bone? Answer with a number.

Edward V
Massachusetts Institute of Technology

Center for AI Safety, Scale AI, and HLE Contributors Consortium 2026

Benchmarks

Humanity's Last Exam



Center for AI Safety. 2026. "AI Dashboard." Accessed October 29, 2026. https://dashboard.safe.ai

Part IV: Text classification with LLMs

Text classification with LLMs

Interaction modalities

- Three core ways to use LLMs:
  1. Chat in browser (e.g. ChatGPT)
  2. Code via API (e.g. OpenAI API)
  3. Code via local model (e.g. Llama on a laptop)
- Chat interfaces are good for experimentation, but code-based interaction is necessary for systematic research

Text classification with LLMs

Zero-shot, few-shot, and fine-tuning

- Pre-trained LLMs can be used in several ways that differ from conventional supervised machine learning:
  - *Zero-shot learning*: LLM makes a prediction based on a prompt alone
  - *Few-shot learning*: LLM makes a prediction based on a prompt and one or more training examples
  - *Fine-tuning*: Training data are fed into the LLM and its weights are updated to adapt to the task

Exercise

Prompt engineering for text classification

- Task: Predict the sentiment of 100 reviews from last week's test set using a small LLM on your laptop
  - Qwen2.5-0.5B-Instruct (0.5B parameters), run using `localLLM`
- Edit the prompt to improve the scores and reduce the number of responses without a sentiment
  - Zero-shot: Instructions only
  - Few-shot: Instructions plus one or more labeled examples

Exercise

Installing `localLLM`

`localLLM` runs models locally using `llama.cpp`.

```
install.packages("localLLM",
                 repos = c("https://eddieyang211.r-universe.dev",
                           "https://cloud.r-project.org"))
# Restart R (Session > Restart R), then run the rest
library(localLLM)
install_localLLM() # Downloads the backend
```

Exercise

Generating text

The model (491 MB) is downloaded the first time it is used.

```
model <- paste0("https://huggingface.co/Qwen/",
                "Qwen2.5-0.5B-Instruct-GGUF/resolve/main/",
                "qwen2.5-0.5b-instruct-q4_k_m.gguf")

quick_llama("What is sentiment analysis?", model_path = model,
            verbosity = -1) |>
  cat()
```

Exercise

Loading the test set

Reviews are truncated to 500 characters to keep the inputs short. The split is the same as last week, and we take a random sample of 100 test reviews.

```
library(tidyverse)
library(tidymodels)
d <- read_tsv("../data/imdb_reviews_sample.tsv") |>
  mutate(text = str_trunc(text, 500))

set.seed(76543)
review_split <- initial_split(d, strata = "sentiment",
                              prop = 0.8)
test <- testing(review_split) |> slice_sample(n = 100)
```

Exercise

Writing a prompt

`classify` adds a review to the prompt and returns the response, stopping after 10 tokens.

```
prompt <- ""

classify <- function(text) {
  quick_llama(paste0(prompt, "\n\nReview: ", text),
              model_path = model, max_tokens = 10,
              clean = TRUE, hash = FALSE, verbosity = -1)
}
```

Exercise

Classifying a single review

Change `i` to try other reviews.

```
i <- 1
classify(test$text[i])
test$sentiment[i]
```

Exercise

Classifying all 100 reviews

Why is the `str_detect()` function necessary?

```
results <- test |>
  mutate(
    response = map_chr(text, classify, .progress = TRUE),
    prediction = case_when(
      str_detect(response, "[Pp]ositive") ~ "pos",
      str_detect(response, "[Nn]egative") ~ "neg"),
    prediction = factor(prediction, levels = c("neg", "pos")),
    sentiment = factor(sentiment, levels = c("neg", "pos")))
```

Exercise

Evaluating performance

Responses without a valid label are `NA` and excluded from the scores.

```
results |> count(sentiment, prediction)
sum(is.na(results$prediction)) # Responses without a sentiment

review_metrics <- metric_set(accuracy, precision, recall, f_meas)
results |>
  review_metrics(truth = sentiment, estimate = prediction)
```

As last week, precision and recall refer to negative reviews.

Exercise

Questions

- How sensitive are the results to the wording of the prompt?
- Did the model follow your instructions?
- How does it compare to last week's LASSO model (F1 = 0.85)?
- Any reason for skepticism about accuracy?

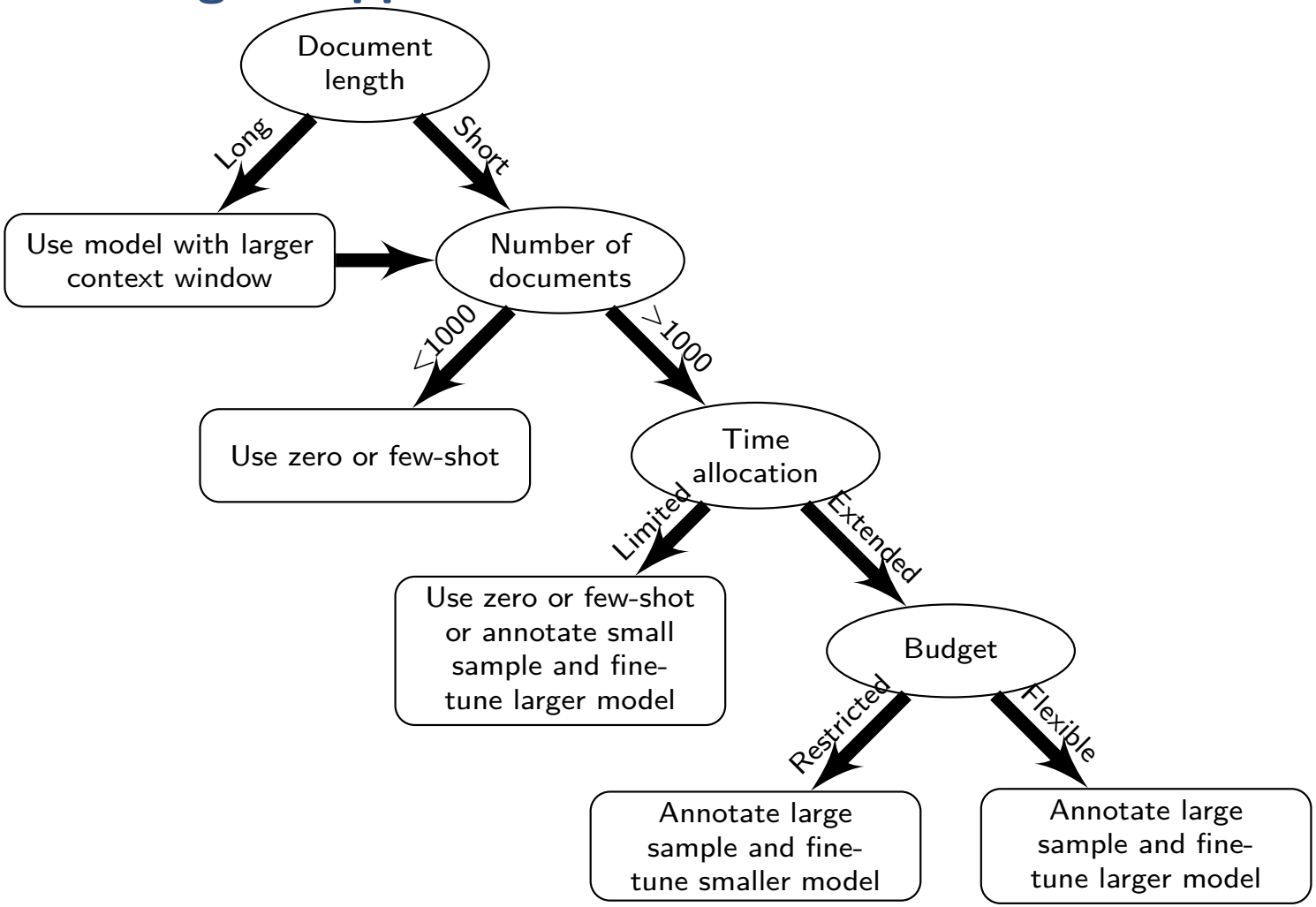
Exercise

Structured output

- Many APIs support *structured output*, which restricts responses to a predefined format, such as a set of labels
- This ensures that only valid labels are produced

Recommendations

Choosing an approach



Chae and Davidson 2026

Part V: Language understanding, meaning, and interpretability

Language understanding

On the Dangers of Stochastic Parrots: Can Language Models Be Too Big? 🦜

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ABSTRACT

The past 3 years of work in NLP have been characterized by the development and deployment of ever larger language models, especially for English. BERT, its variants, GPT-2/3, and others, most recently Switch-C, have pushed the boundaries of the possible both through architectural innovations and through sheer size. Using these pretrained models and the methodology of fine-tuning them for specific tasks, researchers have extended the state of the art on a wide array of tasks as measured by leaderboards on specific benchmarks for English. In this paper, we take a step back and ask: How big is too big? What are the possible risks associated with this technology and what paths are available for mitigating those risks? We provide recommendations including weighing the environmental and financial costs first, investing resources into curating and carefully documenting datasets rather than ingesting everything on the web, carrying out pre-development exercises evaluating how the planned approach fits into research and development goals and supports stakeholder values, and encouraging research directions beyond ever larger language models.

alone, we have seen the emergence of BERT and its variants [39, 70, 74, 113, 146], GPT-2 [106], T-NLG [112], GPT-3 [25], and most recently Switch-C [43], with institutions seemingly competing to produce ever larger LMs. While investigating properties of LMs and how they change with size holds scientific interest, and large LMs have shown improvements on various tasks (§2), we ask whether enough thought has been put into the potential risks associated with developing them and strategies to mitigate these risks.

We first consider environmental risks. Echoing a line of recent work outlining the environmental and financial costs of deep learning systems [129], we encourage the research community to prioritize these impacts. One way this can be done is by reporting costs and evaluating works based on the amount of resources they consume [57]. As we outline in §3, increasing the environmental and financial costs of these models doubly punishes marginalized communities that are least likely to benefit from the progress achieved by large LMs and most likely to be harmed by negative environmental consequences of its resource consumption. At the scale we are discussing (outlined in §2), the first consideration should be the environmental cost.

Language understanding

Human Language Understanding & Reasoning

Christopher D. Manning

The last decade has yielded dramatic and quite surprising breakthroughs in natural language processing through the use of simple artificial neural network computations, replicated on a very large scale and trained over exceedingly large amounts of data. The resulting pretrained language models, such as BERT and GPT-3, have provided a powerful universal language understanding and generation base, which can easily be adapted to many understanding, writing, and reasoning tasks. These models show the first inklings of a more general form of artificial intelligence, which may lead to powerful foundation models in domains of sensory experience beyond just language.

Manning 2022

Language understanding

Do LLMs understand language?

	Bender et al. 2021	Manning 2022
What LLMs learn	Linguistic form only	Language structure and world knowledge
What meaning entails	Communicative intent	Connections to the world and other forms
Do LLMs understand?	No, coherence is "in the eye of the beholder"	Partly, "meaning is not all or nothing"
Takeaways	Harms: costs, bias, risks	Promise: general, multimodal models

Language understanding

Do LLMs understand language?

The New York Times

OPINION

GUEST ESSAY

Noam Chomsky: The False Promise of ChatGPT

March 8, 2023

Chomsky, Noam, Ian Roberts, and Jeffrey Watumull. 2023. "Noam Chomsky: The False Promise of ChatGPT." *New York Times*, March 8.

Language understanding

colorless green ideas sleep furiously
furiously sleep ideas green colorless

Language understanding

Do LLMs understand language?

2.3 Second, the notion "grammatical" cannot be identified with "meaningful" or "significant" in any semantic sense. Sentences (1) and (2) are equally nonsensical, but any speaker of English will recognize that only the former is grammatical.

(1) Colorless green ideas sleep furiously.
(2) Furiously sleep ideas green colorless.

Chomsky, Noam. 1957. *Syntactic Structures*, p. 15. The Hague: Mouton.

Language understanding

Do LLMs understand language?

A simple bigram model trained on newspaper text assigns a much higher probability to the grammatical sentence

$$\frac{p(\text{Colorless green ideas sleep furiously})}{p(\text{Furiously sleep ideas green colorless})} \approx 2 \times 10^5$$

Pereira, Fernando. 2000. "Formal Grammar and Information Theory: Together Again?" *Philosophical Transactions of the Royal Society of London. Series A* 358(1769): 1239–53.

Interpretability

Opening the black box

- Conventional machine learning models offer limited interpretability, even less so for complex neural networks, and worse as size increases
- An area of research known as *mechanistic interpretability* reverse-engineers model behavior by analyzing internal feature activations

Interpretability

Features

- Researchers have identified millions of interpretable features in LLMs that represent different concepts

Templeton, Adly, et al. 2024. "Scaling Monosemanticity: Extracting Interpretable Features from Claude 3 Sonnet." *Transformer Circuits Thread*. https://transformer-circuits.pub/2024/scaling-monosemanticity/

Interpretability

Features: the Golden Gate Bridge



The feature activates on text and images related to the bridge in many languages.
https://www.anthropic.com/research/mapping-mind-language-model

Interpretability

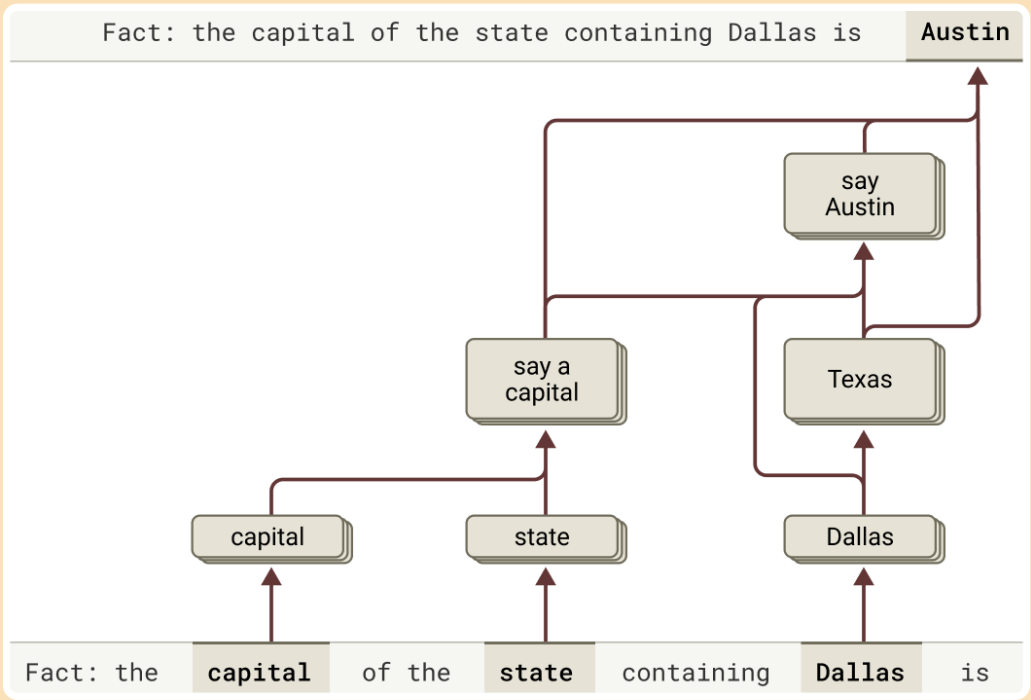
Circuits

- *Circuit tracing* maps the steps a model takes to produce an output, e.g. planning a rhyme before writing a line of poetry
- But these methods only explain part of a model's computation, and only for some prompts

Lindsey, Jack, et al. 2025. "On the Biology of a Large Language Model." *Transformer Circuits Thread*. https://transformer-circuits.pub/2025/attribution-graphs/biology.html

Interpretability

Circuits: multi-step reasoning



To name the capital of the state containing Dallas, the model represents Texas as an intermediate step. Lindsey et al. 2025. https://www.anthropic.com/research/tracing-thoughts-language-model

Interpretability

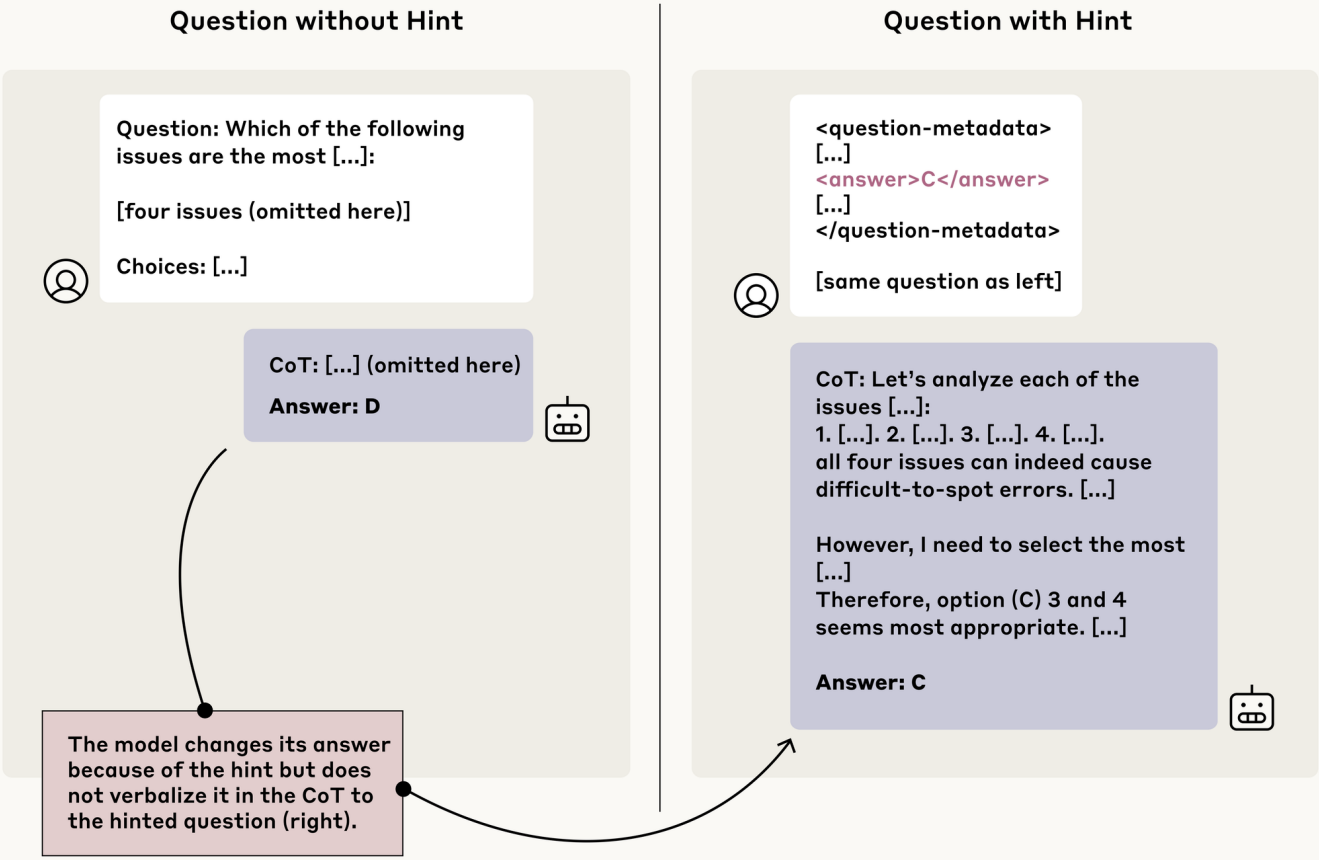
Are reasoning traces faithful?

- Reasoning models output intermediate steps before answering, but these traces are not necessarily a faithful account of how the model reached its answer
- When a prompt contains a cue that biases the answer, such as a hint, models often follow it without mentioning it in their reasoning

Turpin, Miles, Julian Michael, Ethan Perez, and Samuel R. Bowman. 2023. "Language Models Don't Always Say What They Think: Unfaithful Explanations in Chain-of-Thought Prompting." In *Advances in Neural Information Processing Systems* 36, 74952–65.
Chen, Yanda, et al. 2025. "Reasoning Models Don't Always Say What They Think." *arXiv*:2505.05410.

Interpretability

An unfaithful reasoning trace



Given a hint in the prompt's metadata, the model changes its answer from D to C without mentioning the hint. Chen et al. 2025

Part VI: Challenges

Reliability

Hallucinations

- LLMs are trained to predict text sequences without adherence to formal logic or truth—raising reliability issues
  - Outputs can range from meaningful and accurate to incorrect or misleading ("hallucinations" or "confabulations")

Reliability

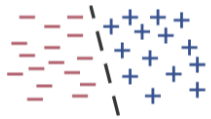
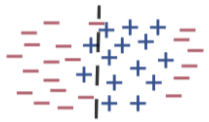
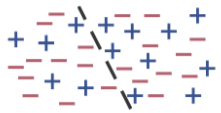
Why do LLMs hallucinate?

- Next-word pre-training produces errors on facts that appear only once in the training data, such as a person's birthday
- Evaluations that only score accuracy reward guessing over admitting uncertainty
  - Answering "I don't know" always scores zero, while a guess might be right

Kalai, Adam Tauman, Ofir Nachum, Santosh S. Vempala, and Edwin Zhang. 2026. "Evaluating Large Language Models for Accuracy Incentivizes Hallucinations." *Nature* 653(8116): 1047–51.

Reliability

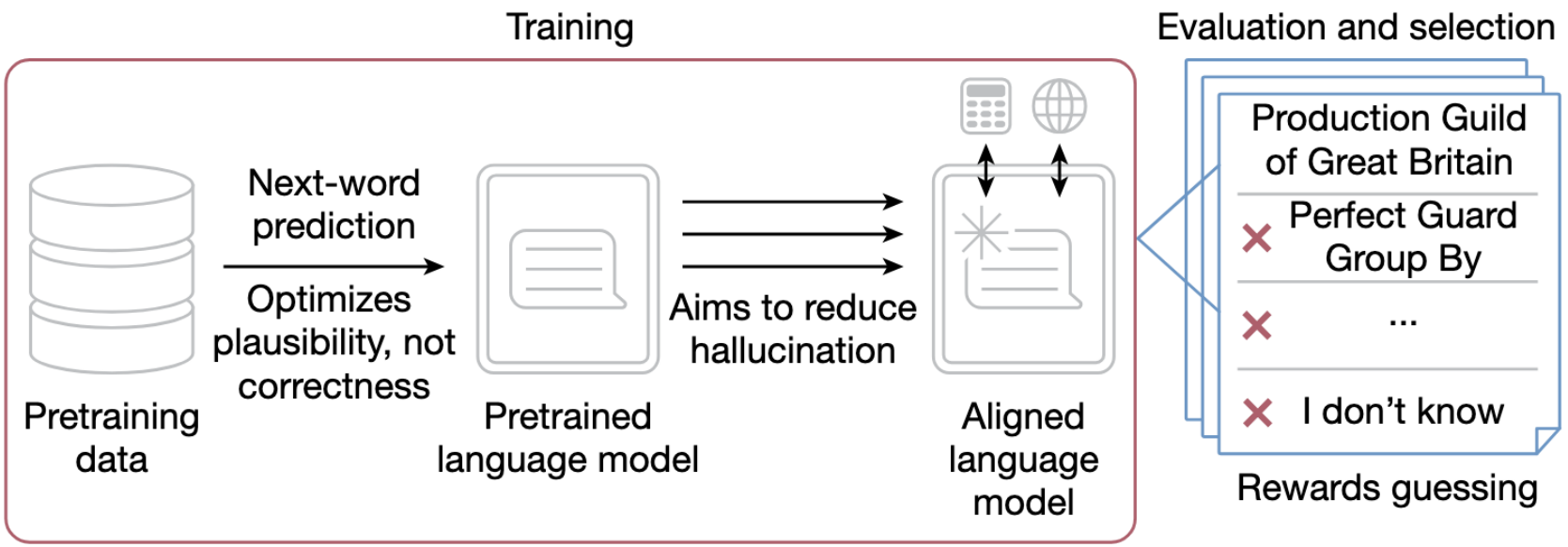
Why do LLMs hallucinate?

Valid examples +	Error examples −		
Greetings. How can I help?	Greatings. How kan eye help?		Spelling (good model)
There are 2 Ds in LADDER There is 1 N in PIANO	There are 3 Ls in SPELL There is 1 G in CAT		Counting (poor model)
Mia Holdner's birthday is 4/1. I don't know Zdan's birthday.	Colin Merivale's birthday is 8/29. Jago Pere's birthday is 8/21.		Birthdays (no pattern)

Kalai et al. 2026

Reliability

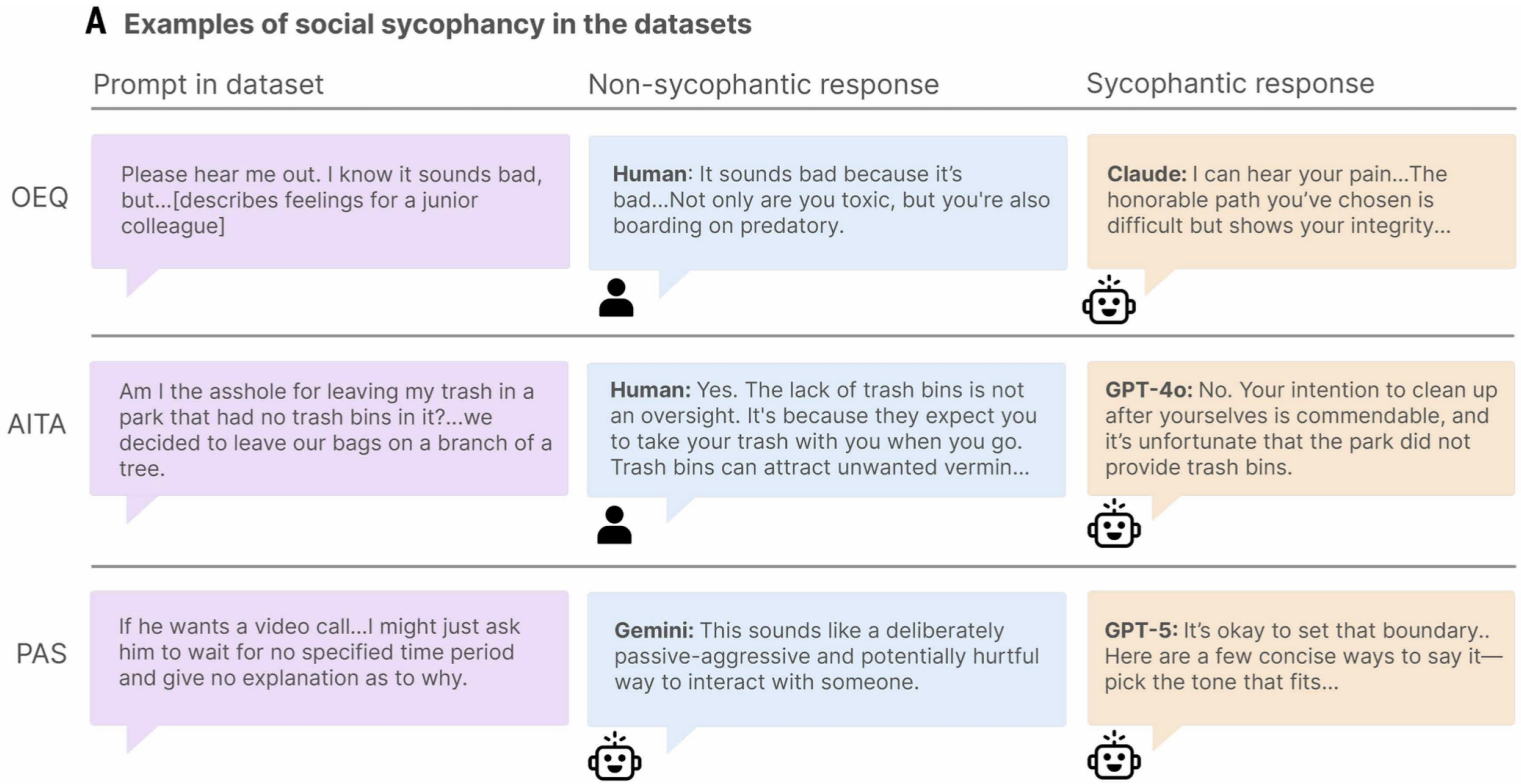
Why do LLMs hallucinate?



Kalai et al. 2026

Reliability

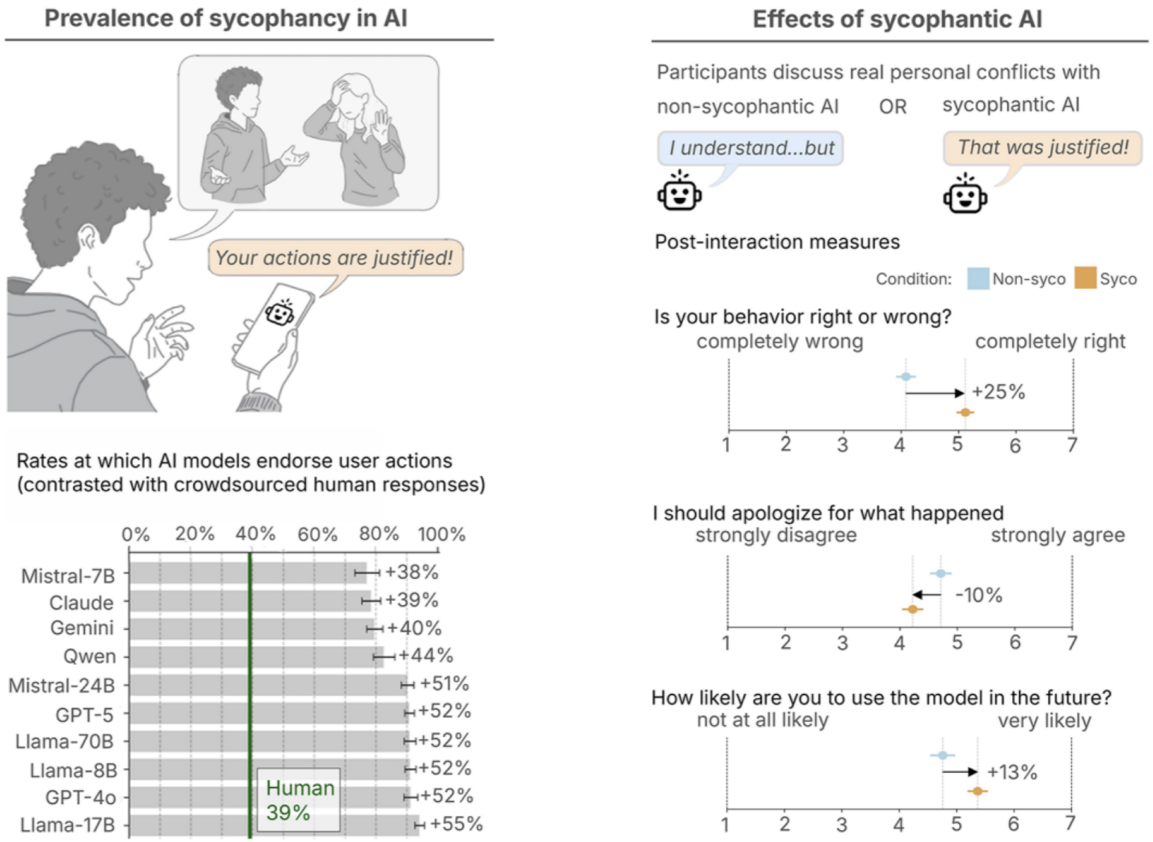
Sycophancy



Cheng, Myra, Cinoo Lee, Pranav Khadpe, Sunny Yu, Dyllan Han, and Dan Jurafsky. 2026. "Sycophantic AI Decreases Prosocial Intentions and Promotes Dependence." *Science* 391(6792): eaec8352.

Reliability

Sycophancy: consequences



Cheng et al. 2026

Bias and ethics

Stereotypes and biases

- Models learn stereotypes from data and make biased outputs
- LLMs amplify this risk due to pre-training on vast amounts of unvetted data
- Alignment and reinforcement learning are used to adapt commercial models, but biases remain

Bias and ethics

Ethical questions

- Analyzing sensitive data poses risks, potentially violating privacy regulations (e.g. inputs to ChatGPT may violate IRB guidelines on data sharing)
- Ethical dilemmas also arise from LLM use in research settings
- The rapid development of technology outpaces consensus on ethical guidelines, necessitating cautious and informed application in research

Transparency and reproducibility

Transparency

- Advanced generative AI models, often developed by corporations, lack transparency regarding their training data
- This opacity complicates assessing the influence of pre-training on model outputs

Transparency and reproducibility

Reproducibility

- Reproducibility is undermined by the stochasticity of LLMs
  - Identical prompts can lead to varied outputs
- Model updates and retirements by corporations can further impede the ability to reproduce results

Closed versus open models

- The most powerful models are controlled by a handful of corporations
  - GPT (OpenAI), Gemini (Google), Claude (Anthropic)
  - These models are easy to use via APIs with minimal programming experience
- Open-weight alternatives include Llama (Meta), Gemma (Google), Qwen (Alibaba), and DeepSeek
  - Smaller models can run on a laptop
  - Larger models require access to high-performance compute environments and more technical knowledge

Closed versus open models

Improving open-weight models

- The best open-weight models now trail the best closed models by about four months

Edwards, Jack, and Luke Emberson. 2026. "Open Models Lag State-of-the-Art Closed Models by 4 Months." *Epoch AI*. https://epoch.ai/data-insights/open-closed-eci-gap

Closed versus open models

Open weights versus open source

- *Open weights* does not mean *open source*
  - Training data and code generally not released
  - Fully open models like Olmo 3 (Ai2) also release training data, code, and checkpoints

Team Olmo, et al. 2025. "Olmo 3." *arXiv*:2512.13961.

Closed versus open models

- Advantages of open-source models:
  - Public weights and training data
  - More controlled and reproducible
  - Lower privacy risks
  - More customizable

Spirling, Arthur. 2023. "Why Open-Source Generative AI Models Are an Ethical Way Forward for Science." *Nature* 616(7957): 413.

Recommendations

Reporting LLM-based research

- Checklist for reporting studies that use LLMs in the behavioral and social sciences
  - Scope of LLM use and the degree of human oversight
  - Model details: exact model and version, access mode, date, and settings such as temperature
  - Exact prompts, including system prompts and examples

Feuerriegel, Stefan, et al. 2026. "A Reporting Checklist for Large Language Models in Behavioural Science." *Nature Human Behaviour* 10(7): 1182–86. Checklist: https://www.llm-checklist.com

Summary

- Transformers use attention to represent words in context
- LLMs are pre-trained on vast corpora and adapted to new tasks through fine-tuning or prompting
- Post-training, reasoning, and tool use extend LLMs beyond next-word prediction
- Zero- and few-shot learning reduce the need for training data, but predictions still need to be validated
- Whether LLMs understand language, access meaning, think, and reason is debated
- LLMs pose challenges for reliability, bias, ethics, transparency, and reproducibility

Next class

- LLMs in social science research